

**CLINICAL
NEUROSCIENCE
PSY/NSC 4119**

**LABORATORY
GUIDE**

LABORATORY IN
CLINICAL NEUROSCIENCE

PSY/NSC 4119

LABORATORY GUIDE

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COLLEGE, 1970 REVISION.

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Lab 1

Neurons and Neuroglia

Introduction: This lab is a review of neuronal and glial cell structure and function. While this will probably be a review for many of you, these are important basic concepts that the rest of the semester builds upon. A brief lecture will be given to cover the material and allow for questions, during the lecture you will be expected to complete the worksheet that follows this page. At the conclusion of the lecture you will be given the opportunity to observe some of the structures under a microscope. By the end of the class, you should be able to identify and define the key terms.

Key Terms / Structures:

Afferent neuron	Glial cells	Pseudounipolar neuron
Astrocytes	Grey matter	Pyramidal neuron
Axoaxonal synapse	Interneuron	Schwann cell
Axodendritic synapse	Multipolar neuron	Synapse
Axosomatic synapse	Myelin	Unipolar neuron
Bipolar neuron	Neuron	White matter
Dendrites	Nodes of Ranvier	
Efferent neuron	Oligodendrocyte	

1. Nerve Cells

Grey matter – cell bodies

- outer surface of cortex
- internal brain structures (_____ in CNS)
- peripheral (_____ in PNS)

White matter – myelinated axons

- Bundles of myelinated axons:
- (_____ in CNS)
- (_____ in PNS)

Neuron – an individual nerve cell

down axon : _____
across synapse : _____

Neuronal Cell Parts

Dendrites – (1°, 2°, 3°)

Cell body or soma –

Axon –

Axon Terminal –
Axon Collaterals –
Synaptic Vesicles –

Neuronal Cell Parts (continued)

Receptors:

Synapse:

Myelin:

Nodes of Ranvier

2. Complexity – The Big Picture

The way the neurons are connected gives the nervous system its complexity of function.

A neuron may receive signals on:

These signals may be _____ or _____ in nature.

Types of Synapses

- 1.
- 2.
- 3.

3. Classification of Neurons

Classification by Shape

unipolar

pseudounipolar

bipolar

multipolar

pyramidal (apical and basilar dendrites)

Classification by Function

afferent: (sensory)

efferent: (motor)

interneuron: Connector

Reflex arc example

4. Glial Cells

Function: structural and metabolic support

- 1.
- 2.
- 3.

Types:

Schwan Cells – myelin in _____

Oligodendrocyte – myelin in _____

Astrocyte – packing, support, physiology

Fibrous

Protoplasmic

Satellite

Name _____

Homework:

1. Draw a neuron and label the following: *Dendrites, Axon Terminal, Node(s) of Ranvier, Axon, Myelin, Nucleus, Soma, Terminal boutons (nerve terminal, axon terminal)*
2. Draw 3 synapses (on neuron #1 above) representative of the synapse types discussed in lab.
3. Draw a monosynaptic reflex arc.

Homework continuation page

Lab 2

Development of the Nervous System

Introduction: This lab is an overview of the development of the central nervous system (CNS). We will track development of the brain and spinal cord from zygote to birth, and discuss what can go wrong and the resulting clinical abnormalities. A brief lecture covers basic concepts and allows for questions. During the lecture you will be expected to complete the worksheet that follows this page. There are five major divisions of the brain: the telencephalon, diencephalon, mesencephalon, metencephalon, and myelencephalon. During this lab you will learn where these different regions are and some their main functions.

At the conclusion of the lecture you will be given a chance to put away all your class materials and get out your lab materials (gloves, brain bucket, & safety glasses) and we will discuss lab safety / policies and how to handle the brains. You will then be given your brain and some time to begin the process of cleaning the pia mater from the brain. We will continue the cleaning process in the next lab. By the end of the class you should be able to identify and define the following key terms.

Key Terms / Structures:

Alar plate	Mesoderm	Primitive node
Anencephaly	Morula	Primitive streak
Basal plate	Neural crest cells	Somites
Blastocyst /	Neural fold	Spina bifida
Blastula	Neural groove	Tropoblasts
Ectoderm	Neural plate	Zygote
Embryonic disk	Neural pore	
Endoderm	Neural tube	
Induction	Neurulation	
Inner cell mass	Notocord	

CNS Development

Formation of Layers

Overview

- Zygote: fertilized egg (cells double in number every 24 hrs)
- Morula stage (72 hrs)
- Blastocyst – hollow ball
 - outer cells
 - inner cells
- Embryonic disk (germinal disk)
 - Two original layers
 - Epiblast-
 - Endoblast-
 - Primitive pit
 - Primitive Streak
 - Neural plate

Three Principal Layers

1. Ectoderm
2. Mesoderm
3. Endoderm

Formation of the Nervous System

Induction

Mesoderm will form _____ and is required for _____
Ectoderm **with** induction becomes:
Ectoderm **without** induction becomes:

Two mesoderm cell groups

Notocord –

Somites –

Neural Plate:

Two lines of development from neural plate

- 1.
- 2.

Neurulation

Definition:

Neural Folds:

Neural Groove:

Neural Crest:

Neuropores:

Anencephaly:

Spina Bifida:

Neural ectoderm:

Three Layers

Marginal Layer – outer →

Mantle Layer – middle →

Matrix Layer – Ependyma – inner →

Three Cell Types

Neuroblast:

Glioblasts:

Ependyma:

Spinal Cord

Alar Plate: (located)_____

Basal Plate: (located)_____

Neural Crest Cells

Migrate away from neural tube and become PNS structures

All ganglia (see below)

Dorsal Root Ganglia (DRG)

Peripheral Glial Cells

Sympathetic Ganglia

Parasympathetic Ganglia

Adrenal medulla cells

Melanocytes

Inner meningeal layers:

Directional Terms

Midsagittal Plane

Right & Left

Medial

Lateral

Ipsilateral

Contralateral

Coronal Plane

Anterior (frontal or ventral – belly)

Posterior (back or dorsal)

Horizontal (Transverse, Axial) Plane

Superior

Inferior

Limbs

Proximal

Distal

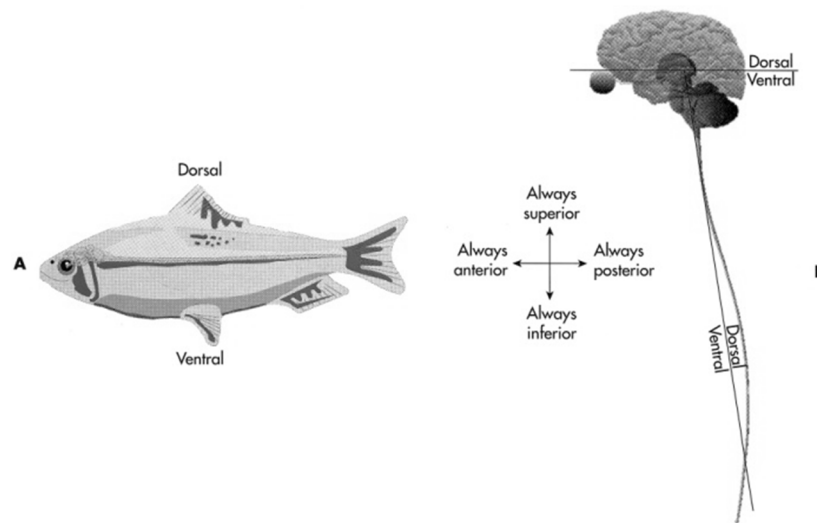
Others

Rostral

Caudal

The terms anterior/posterior and superior/inferior are **ABSOLUTE**. Because of our upright posture, and the bend in the neuraxis produced by the cephalic flexure (located in the mid-brain, see below), the terms dorsal/ventral are **RELATIVE** to the rostro-caudal location.

- Caudal to the cephalic flexure (pons, medulla, spinal cord): dorsal and posterior are synonymous; ventral and anterior are synonymous; rostral is superior and caudal is inferior.
- Rostral to the cephalic flexure (diencephalon, telencephalon): dorsal and superior are synonymous; ventral and inferior are synonymous; rostral is anterior and caudal is posterior.



CNS Divisions

1. Spinal Cord

dorsal –
ventral –

2. Myelencephalon

____ ventricle
Nerves:

3. Metencephalon

____ ventricle
Nerves:

4. Mesencephalon

Aqueduct of Sylvius or _____
Nerves:

5. Diencephalon

____ ventricle
Nerve:

6. Telencephalon

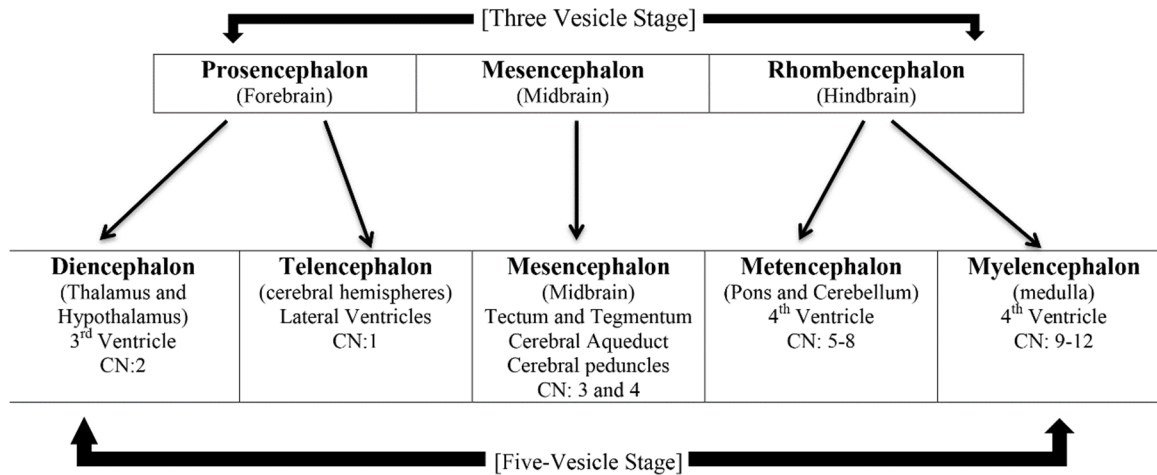
_____ ventricles
Nerve:

CNS Division

Spinal Cord:

- Dorsal – sensory, afferent, towards CNS
- Ventral – motor; efferent, away from CNS (towards periphery)

Vesicle Stages of Development



Procedure

Today, you will familiarize yourself with the instructions for brain dissection. Once you have read Introductory Remarks and Terminology, you will be ready to begin working on your brain. In order to view the structures of the brain, you will need to carefully remove the pia mater and blood vessels.

1. Dissection Guide

Introductory Remarks

The following instructions are a guide to help you dissect the brain in an orderly manner. They are not intended to be complete or absolute and additions to the required steps can and should be made to accord with personal needs so that a clear picture of the various structural relationships is obtained.

You should always proceed slowly in your dissection so as not to prematurely destroy structures. You should carry out as much of the dissection as possible on one half of the brain so that any prematurely destroyed structure can later be recognized on the other half. In general, one half of the brain will be used for horizontal sections and the other for coronal sections.

The brain of the sheep is particularly adaptable for elementary study because it is almost identical in structure with the brain of higher animals and it is, therefore, an introduction to the more developed, but essentially similar, brain of man. In size and

form, it ranks near the upper end of the vertebrate series, and presents all the important constituents of the central organ of the nervous system in its highest development.

*You should complete the assigned drawings of the sheep brain from ventral (V), lateral (L), dorsal (D), and sagittal (S) views, as instructed. You are to (1) locate on your preparation the features outlined in these instructions **and** listed in the "Key Structures;" (2) label each feature on the drawings.*

Although a given structure may appear most prominently from one view, please label each feature in all drawings in which it appears (e.g., the optic chiasm should be labeled on the ventral, sagittal, and lateral views of the brain). Be sure the labels are legible and complete. Place the labels outside and around the drawings. Use a ruler to draw straight lines from the labels to the structures. Do not clutter the drawings—use as many drawings of the same view as necessary to clearly indicate all features appearing in that view. Certain features to be labeled do not appear with full clarity in the drawings. In such cases add detail if necessary or point out the approximate position of the structure. After each structure listed in the "Key Structures," place the appropriate letter to indicate the drawings on which you located the structure (e.g. optic chiasm, V, L, S). When more than one drawing of a single view has been used, number the drawings and indicate the appropriate drawings in the "Key Structures" (e.g. V1, V2, L1, L2, etc.). The "Key Structures" should serve as a complete index to your drawings and will be graded accordingly.

Terminology

Study and be sure you clearly understand the following items of orientation in reference to the central nervous system: dorsal, ventral, anterior, posterior, superior, inferior, proximal, distal, lateral, medial, rostral, caudal, and cranial. Also be sure you know the type of brain section indicated by (1) sagittal or longitudinal (special case of midsagittal or median), (2) coronal or frontal, (3) horizontal, and (4) transverse. Confusion arises with these orientational terms because man and other primates walk on their hind legs, with their spinal cord entering at the back of the brain.

The names of anatomical structures are usually descriptive Latin terms. Some authors use the Latin terminology while others use the anglicized form of the Latin names (e.g. the Latin corpus trapezoidem becomes the trapezoid body; capsula interna becomes the internal capsule). This can obviously lead to confusion if you are not alert to the problem. You should also study the English translation of anatomical terms throughout the term. This will often assist your recall of structure. Such translations can be found in any medical dictionary.

To further complicate the terminology picture, the same anatomical structure may have more than one distinct name (e.g. Ammon's horn = hippocampus; cerebral peduncles = crus cerebri), and it may not always be generally agreed how inclusive or exclusive a given name is. You should be alert to this problem. All the common names of a structure will be indicated to you whenever possible.

The names of the nuclei (a nucleus is a group of cell bodies within the central nervous system [CNS]; outside the CNS such a group is called a ganglion) within the brain sometimes follow a logical, directional pattern, e.g., lateral geniculate (geniculum, diminutive of genu, knee; thus little knee). Others are purely descriptive, e.g., nucleus gracilis (slender nucleus), hippocampus (sea horse), pulvinar (a couch), and the description is often quite fanciful.

Terminology for the fiber tracts often follows a from-to pattern, e.g., spinothalamic tract (running from the spinal cord to the thalamus), corticospinal (from cortex to spinal cord), spinocerebellar (from spinal cord to cerebellum), reticulospinal (from reticular formation to spinal cord), tectospinal (from tectum to spinal cord), and mammillothalamic (from mammillary bodies to thalamus). However, even here, purely descriptive and historically determined terminology creeps in, e.g. diagonal band of Broca and tract of Vicq d'Azyr. Fiber tracts go by several names, other than tract, which you should recognize. A tract is often called a fasciculus (little bundle) or a stria (narrow band). A funiculus (little cord) is usually larger and less functionally specific than a fasciculus or stria.

Attention to various prefixes will often clarify a seemingly complex name as in periaqueductal gray (around the aqueduct), periamygdaloid area (around the amygdala), interpeduncular cistern (between the peduncles), interventricular foramen (between the ventricles), parafascicularis (beside the fasciculus), pretectal area (in front of the tectum), and retrosplenial (behind the splenium).

2. Remove pia mater from the dorsal surface.

Begin by very carefully picking off pia mater, using a pair of surgical tweezers only on the **dorsal surface ONLY** as to not damage any ventral structures. Your lab instructor will guide you through this process. Next week we will remove the pia mater from the ventral surface.

3. Cleaning

Before leaving lab you must properly and carefully store your brain, clean any equipment you used today, and clean your table.

Name _____

Homework

Draw the following stages of the developing brain. Label prominent structures on your drawings

1. 20 days of development
 - a. dorsal view
 - b. cross-section
2. 22 days of development
 - a. dorsal view
 - b. cross-section
3. 4 weeks of development (3 vesicle stage)
4. 6 weeks of development (5 vesicle stage)

Homework continuation page

Lab 3

Cranial Nerves; External Anatomy

The Cranial Nerves

Introduction: Learning the cranial nerves may seem daunting at first, but don't be intimidated. There is a very methodical organization to these tracts that exchange afferent and efferent information with the head, neck, and visceral organs. They can generally be divided into categories of somatic sensory, visceral sensory, somatic motor, and visceral motor. Some are a hybrid of these divisions, relaying sensory information from one part of the body and sending motor commands to another. Senses like taste and hearing are transmitted via the special sensory nerves. The cranial nerves that innervate structures derived from the branchial (or gill) arches (muscles of the face, jaw, larynx, and pharynx) are called the *branchial* nerves. As you study each of the twelve nerves, it is helpful to group them together in terms of their sensory or motor functions. They are numbered in ascending manner from rostral to caudal, and from medial to lateral.

Categories of Nerve

Sensory

Somatic (SS)

Visceral (VS)

Special (SP)

Motor

Somatic (SM)

Visceral (VM)

Branchial (BM)

Rhyme	#	Name	M / S	Structure Innervate	Function
	1				
	2				
	3				
	4				
	5				
	6				

7	8	9	10	11	12

External Anatomy

Introduction: Be able to identify the following structures on the sheep brain specimens. There will be structures discussed in the dissection guide and on the list at the beginning of this section that will not be in the sheep brain atlas, you are still responsible for ALL structures. Be sure to find all the cranial nerves on your own sheep brain and identify their function. At first, this may seem an overwhelming task, but don't despair. Over the course of the semester, we will return to many of these structures.

DO NOT feel intimidated by this long list of structures. The purpose of today's lab is to begin to obtain a sense of familiarity with several structures. As we proceed through the lab course, these structures will arise again. As you proceed in the lecture course, you will hear many of these terms again. Together, this will facilitate your understanding of basic brain anatomy and function.

Key Structures:

Abducens nerve	Hypothalamus
Ansate sulcus and cruciate fissure	Inferior colliculus
Anterior perforated substance	Infundibulum
Brachium pontis (middle cerebellar peduncle)	Insula
Brain stem	Interpeduncular cistern
Cerebellum	Lateral and medial olfactory stria
Cerebral cortex	Mammillary body
Cerebral hemispheres	Medial longitudinal (interhemispheric) fissure
Cerebral peduncles	Medulla oblongata
Cisterna magna	Occipital pole
Corpora quadrigemina	Oculomotor nerve
Cuneate tubercle	Olfactory bulb
Diagonal band of Broca	Olfactory nerve
Dorsal median fissure	Olfactory sulcus
Facial nerve	Olive
Fasiculus cuneatus	Optic chiasm
Fasiculus gracilis	Optic tract
Foramen of Lushka	Orbital gyrus
Foramen of Magendie	Parahippocampal gyrus
Glossopharyngeal nerve	Parahippocampal gyrus (Pyriform lobe)
Gracile tubercle	Pituitary gland
Gyrus (gyri)	Pons
Gyrus rectus	Postcentral gyrus
Hypoglossal nerve	Precentral gyrus

Pyramids
Rhinal fissure
Rhinencephalon
Spinal accessory nerve
Sulcus (sulci)
Superior cistern
Superior colliculus
Superior frontal sulcus

Sylvian fissure
Tractus lateralis minor
Trapezoid body
Trigeminal nerve
Trochlear nerve
Vagus nerve
Ventral median fissure
Vestibulo-cochlear nerve

Procedure

1. Removing Pia Mater

After you have identified all the cranial nerves, pick the rest of the pia matter from the ventral surface of the sheep brain, again being careful not to damage any other structures

2. Dissection Guide

External Features of the Brain

Of the three coverings of the brain, your specimen most probably will have only the pia mater completely intact. The dura mater has been removed with the skull. The arachnoid, a rather filmy membrane lying between pia and dura, will be found relatively intact in protected areas. You should study the relative positions of the meninges from your text. As you proceed in your study of the brain, you should carefully clean the pia and blood vessels from the brain. Always do this slowly as you go so that you will not destroy structures. You must be particularly careful not to pull off cranial nerves with the pia.

On the whole brain, locate the following major subdivisions of the central nervous system. The myelencephalon, or the **medulla oblongata**, is the most caudal portion of the brain. It may be distinguished from the metencephalon, which is located immediately rostral of it, by the prominent swelling associated with the latter. This swelling is called the **pons** and consists of fibers that are passing dorsally to reach a second part of the metencephalon, or **cerebellum**. Rostral to the metencephalon is the mesencephalon, or midbrain. When this is viewed on the ventral surface of the brain, it is distinguished by two columns, separated by a depression, extending between the metencephalon and the posterior portion of the next brain division, the diencephalon. These columns, the cerebral peduncles, are composed of myelinated fibers, which travel from the cerebral cortex to the pons and spinal cord. To study these ventral structures you must remove the **pituitary gland**. First, study its attachment to the brain and then remove it being careful not to tear of the fifth (**trigeminal**) nerve, the large trunks of which straddle the gland. Try to do as little damage as possible to the stalk of the pituitary.

On the dorsal aspect of the midbrain are four elevations, the **superior and inferior colliculi**, collectively called the **corpora quadrigemina**. These may be seen if the cerebellum is gently separated from the cerebral hemispheres. Be careful not to tear the connection between the cerebellum and the pons when you are making this observation. Rostral to the mesencephalon is the diencephalon. The portion of the diencephalon, which you will see on the ventral surface of the brain, is the **hypothalamus**. The thalamus, which is another portion of the diencephalon, will be seen when sagittal sections of the brain are studied. The diencephalon is bordered posteriorly by a well-marked prominence, the **mammillary body**, which in man is divided by a longitudinal crease into two mammillary bodies. Anteriorly, the **optic chiasm** marks the rostral border of the diencephalon. The remainder of the brain, i.e. the **cerebral hemispheres**, constitutes the telencephalon. The cerebral hemispheres include the **cerebral cortex** (which refers only to the outer layer of gray matter, cortex = bark of a tree), the underlying white matter of fibers, various nuclei within the hemispheres, and the **rhinencephalon** ("smell brain"). The rhinencephalon includes all those structures directly related to olfaction (**olfactory bulb, nerve, and tract** and the **anterior perforated substance**) and others that are not related but are structurally associated (septal area, hippocampus, and **parahippocampal gyrus**). These five divisions, together with the spinal cord, constitute the central nervous system. The accompanying Table 1 (page 108) outlines the major divisions and subdivisions of the brain and their major associated nerves and cavities. Within these five major subdivisions, locate these following structures:

A. Myelencephalon.

On the ventral surface, locate the **pyramids** on either side of the midline. These tracts are made up of fibers coming from the motor regions of the cerebral cortex and passing to the spinal cord. In the mesencephalon, they constitute the **cerebral peduncles**. In the region of the caudal medulla, 80 - 90% (in man) of these pyramidal fibers cross from one side to the other, or decussate. Thus, a lesion to this tract above the level of the medulla would result in signs and symptoms of a motor nature on the contralateral side of the body.

Immediately below the pons is a transverse stria, the **trapezoid body**, which extends across the ventral surface of the medulla. It carries secondary auditory fibers from the cochlear nuclei across the midline to ascend in the contralateral lateral lemniscus. In man, the much larger pons covers this band from view. Just caudal to the trapezoid body is the **tractus lateralis minor**. It passes caudally along the outer border of the pyramids. This tract does not appear as a distinct bundle in man. Lying just ventral to the tractus lateralis minor is a bulge, the **olive**, made by the underlying inferior olivary nucleus, a part of the reticular system receiving fibers from and contributing fibers to the spinal cord and cerebellum. Separating the pyramids in the midline is the **ventral (or anterior) median fissure**.

The dorsal surface of the medulla is largely concealed by the **cerebellum**. By lifting the cerebellum away from the medulla, you can identify the **fasciculus gracilis** (and its rostral extension called the **clava**) and the more laterally placed **fasciculus cuneatus**, longitudinal fiber tracts moving from spinal cord to thalamus. Lateral to cuneatus you will find the **tuberculum cinereum** marking the spinal nucleus and tract of

the fifth cranial nerve (trigeminal). The intersection of the lateral walls of the fourth ventricle is called the **obex**. Also, identify the **dorsal** (or posterior) **median fissure**, separating the gracile fasciculi in the midline. It may be difficult to visualize these dorsal myelencephalon structures without disrupting the **cisterna magna**. If so, delay your study of them until you have seen the metencephalic structures indicated below.

In the medulla, also identify the ninth (**glossopharyngeal**), tenth (**vagus**), eleventh (**spinal accessory**), and twelfth (**hypoglossal**) cranial nerves. The accompanying table 2 (page 109) outlines the names and functions of the 12 cranial nerves. You should handle the identification of the nerves just as you do all the other structures.

B. Metencephalon.

Locate the **brachium pontis** (middle cerebellar peduncle), which, as mentioned before, is the bundle of fibers responsible for the enlarged size of this portion of the brain. Fibers of the brachium pontis arise from cells in the pons and pass to the cerebellum on the other side.

On the ventral surface, locate the sixth (**abducens**), seventh (**facial**), and eighth (**vestibulo-cochlear**) nerves and more rostrally the fifth (**trigeminal**) nerves. At the lateral junction of the medulla and cerebellum, you will notice a little tuft of choroid plexus, which projects from a small aperture in the pial membrane. This is the location of the **foramen of Lushka** and is one of the means by which the cerebro-spinal fluid passes from the **fourth ventricle** in the sub-arachnoid space between the cerebellum and medulla. (The fourth ventricle will be seen in sagittal section although you will see its floor when the cisterna magna is removed.) This enlargement of the sub-arachnoid space is the cisterna magna, also called the cisterna cerebello medullaris. On the dorsal surface of the metencephalon, note the cerebellum and identify the cisterna magna. There is also a medial foramen in this region serving as a communication between the fourth ventricle and the cisterna magna. It is called the **foramen of Magendie**. This foramen will be better seen after the sagittal section.

C. Mesencephalon.

On the ventral surface of the mesencephalon, locate the **cerebral peduncles** and the **interpeduncular cistern**, another reservoir of the cerebrospinal fluid. Dorsally, you may see, once again, the **superior** and **inferior colliculi**. The superior colliculi are much larger than the inferior; much more so in sheep than in man. Also, note the **superior cistern** between the rostro-ventral portion of the cerebellum and the colliculi. (Knowledge of the normal anatomy of these cisternae as outlined in air or contrast media radiographic studies is often helpful in the diagnosis of tumors or other masses in the CNS.) In this region of the brain, locate the third (**oculomotor**) and fourth (**trochlear**) **cranial nerves**. Note that the trochlear is unique among cranial nerves in that it exits dorsally from the brain stem.

D. Diencephalon.

On the ventral surface of this region, locate the optic tract and optic nerve and note again the **optic chiasm** and the **mammillary body**. Between these structures, you will notice the stalk of the **pituitary gland** (hypophysis). This stalk, the **infundibulum**, arises from a small mound or elevation on the ventral surface of the brain underneath the hypothalamus. This area is bounded anteriorly by the optic chiasm and posteriorly by the mammillary bodies. After turning the brain over you may see the pineal body by sighting above and between the superior colliculi after gently separating the cerebral hemispheres. The myelencephalon, metencephalon, mesencephalon, and diencephalon are spoken of collectively as the **brain stem**. The bulbar region of the brain stem generally refers to the myelencephalon although some authors include the pons under this descriptive term.

E. Telencephalon.

The **cerebral cortex** is has many folds or convolutions, separated by grooves. The deeper grooves are called fissures, and mark off the larger brain divisions, called lobes. Smaller grooves are called **sulci** (singular = sulcus) and mark the smaller divisions, called **gyri** (singular = gyrus). Fissure and sulcus, however, are often used interchangeably.

The telencephalon is the most variable area of the brain from man to other animals. The **cerebral cortex** in man is greatly enlarged, burying many of the structures seen on the surface in the sheep's brain. The brain stem, on the other hand, is relatively constant throughout the mammalian class.

Note the cerebral cortex with its many gyri and sulci or fissures. On the dorsal surface, the **medial longitudinal** (interhemispheric) **fissure** separates the two hemispheres. The separation is complete at the anterior end (frontal pole) and posterior end (occipital pole).

If you carefully separate the two hemispheres at the longitudinal fissure, you can visualize the dorsal surface of the **corpus callosum. The **callosal sulcus** separates the corpus callosum from the adjacent midline cortex, the **cingulate gyrus**. The **cingulate sulcus** separates the cingulate gyrus from the more superior gyri. The medial surface of the cortex will be better visualized and more carefully studied after sagittal section of the brain. For now, visualize these structures in relation to the dorsal cortex.**

The **cruciate fissure** intersects the longitudinal fissure to mark off the anterior third of the cortex. The prominence and extent of this fissure varies greatly in different mammals. It is fairly constant in the higher forms and is analogous to the fissure of Rolando (central sulcus or fissure) in man. The gyrus immediately anterior to the cruciate fissure is the **precentral gyrus**, the primary motor area of the cortex. Immediately posterior to the cruciate fissure is the **postcentral gyrus**, the primary somatic sensory area of the cortex. Intersecting the cruciate fissure at right angles and extending anteriorly is the **superior frontal sulcus**. On the lateral surface, locate the **insula** or island of Reil as the floor of a slight depression of the cortical surface. Sensory and motor representation of the viscera are located here and perhaps some of the cortical representation for taste. Arching upward in a dorsal-anterior direction from the insula is the **Sylvian fissure** or lateral fissure or sulcus. The **rhinal fissure** extends horizontally on the line where the lateral and ventral surfaces meet. It separates the neocortex of the

cerebral hemispheres form the rhinencephalon, the phylogenetically older cortex associated with olfactory sense. In man, the development of the neocortex has completely buried the insula and the rhinal fissure and associated rhinencephalic structures.

There are many other sulci and gyri to be seen on the surface of the sheep brain, but they are neither constant in position for the sheep nor homologous to any similar structure in other species.

After you have located the primary fissures, you can delineate the lobes of the cortex. The portion of the cortex anterior to the cruciate fissure is the frontal lobe. The most posterior part of the cortex is the occipital lobe. Between the occipital lobe and cruciate fissure lies the parietal lobe. There is no clear line of demarcation between parietal and occipital lobes. Posterior to the Sylvian fissure and dorsal to the rhinal fissure lies the temporal lobe. Moreover, there is no clear line of demarcation between temporal and occipital lobes.

On the ventral surface, the **olfactory bulb** is easily identified as a pad-like structure placed against the frontal pole. It lies in a depression, the **olfactory sulcus**. The ventral cortex lateral to the sulcus is the **orbital gyrus**. Cortex medial to the sulcus is the **gyrus rectus**. The **lateral** and **medial olfactory stria** course backward from the olfactory bulb. An intermediate olfactory stria can sometimes be seen between the lateral and medial stria as the bulb is lifted slightly. Between the olfactory stria and the optic tract and chiasm is the **anterior perforated substance**, which receives fibers from the medial and lateral olfactory tract and relays to the hypothalamus. Immediately in front of the optic tracts lies a band of lighter color, the **diagonal band of Broca**, which carries fibers from the septal area to the amygdala. Following the lateral line of the rhinal fissure, you will find the **parahippocampal gyrus**. Terminology here is rather confusing. The parahippocampal gyrus is often used synonymously with **pyriform lobe** or area. Generally, the pyriform area refers to the anterior part of the lobe and the lateral olfactory stria. Prepyriform area refers to part of the lateral olfactory stria and the anterior portion of the pyriform area. The posterior part of the hippocampal gyrus is called the entorhinal cortex or area. The medial bulge on the hippocampal gyrus in the vicinity of the optic tract is caused by the underlying amygdaloid nuclei.

3. Cleaning

Before leaving lab you must properly and carefully store your brain, clean any equipment you used today, and clean your table.

Name _____

Homework

1. Draw the sheep brain from each of the following aspects
 - a. Dorsal (Superior)
 - b. Lateral
 - c. Ventral (Inferior)
2. On each drawing, identify the key structures

Homework continuation page

Lab 4

Cerebrospinal Fluid and Ventricles; Mid-Sagittal Structures

Introduction: The brain and spinal cord are encased in the bony structure of the skull and covered by three protective membranes: the dura mater, the arachnoid, and the pia mater. The central nervous system (CNS) is additionally protected by cerebrospinal fluid (CSF) which fills the cerebral ventricles of the brain, the central canal of the spinal cord, and the subarachnoid space surrounding the entire CNS. In addition to learning about CSF this week, you will also be examining the midline structures of your sheep brain specimen by making a mid-sagittal section. The following is a list the major structures you will need to be able to identify on a mid-sagittal section.

Key Structures:

3rd Ventricle	Foramina of Luschka	Pineal body
4th Ventricle	Genu of the corpus callosum	Pons
Anterior commissure	Habenula	Posterior commissure
Anterior medullary velum	Hypothalamus	Retrosplenial cortex
Body of the fornix	Inferior colliculus	Rostrum of the corpus callosum
Callosal sulcus	Inter-ventricular foramen	Septal area
Central canal	Lamina terminalis	Septum pellucidum
Cerebral aqueduct	Lateral ventricle	Splenium of the corpus callosum
Cingulate gyrus	Mamillary body	Subcallosal gyrus
Cingulate sulcus	Marginal gyrus	Superior cistern
Cisterna Magna	Massa intermedia	Superior colliculus
Column of the fornix	Medulla	Tectum
Corpus callosum	Optic chiasm	Tegmentum
Diagonal band of Broca	Paraolfactory gyrus	Thalamus
Foramen of Magendie		

Cerebrospinal Fluid and the Ventricles

1. Protection and Support for the Brain

Dura Mater

Arachnoid

Pia Mater

2. Functions

3. Daily Production

4. Circulation / Pathway

Choroid Plexus

Ventricles

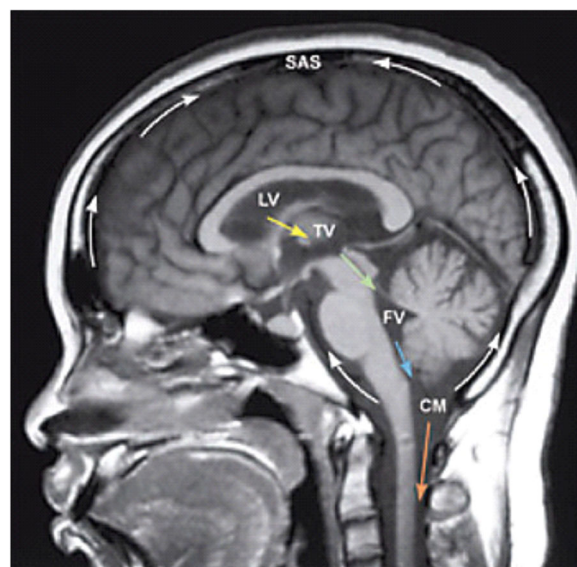
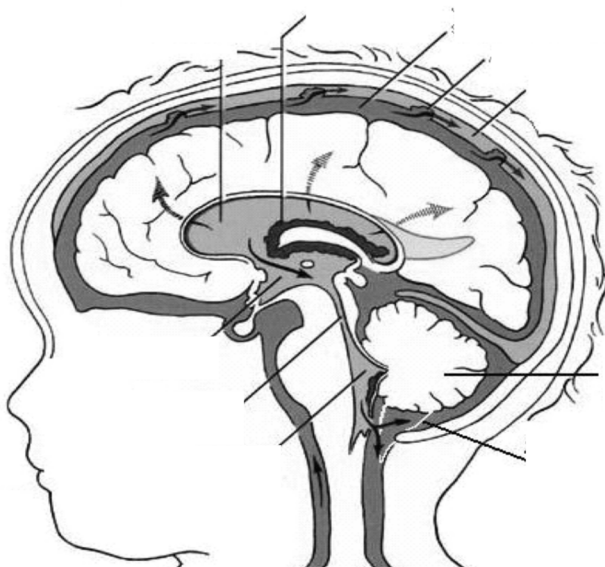
Foramen etc.

Subarachnoid Space

Cisterns

Arachnoid Villa

5. Circumventricular organs



Procedure

To view the mid-sagittal structures of the sheep brain, you will need to cut your sheep brain in half, along the medial longitudinal fissure. Make sure that all of the dura and pia mater have been removed and place the sheep brain on a cutting board, dorsal side up. Using a knife provided by your lab instructor, situate the knife inside the medial longitudinal fissure and apply gentle downward pressure. NOTE: There may be some pia mater remaining on the superior aspect of the corpus callosum and the adjacent cortex deep in the fissure. Do not cut back and forth like a saw; this will damage medial structures. If you have any questions or are uncertain about this process, alert your lab instructor at once, it is imperative that this be a good cut to ensure that you can identify all the necessary structures.

- 1. Removing Pia Mater**

Pick the rest of the pia mater from the ventral surface of the sheep brain, again being careful not to damage any other structures.

- 2. Dissection Guide**

Mid-sagittal Section of the Brain

After covering all of the preceding material on the whole brain, cut the brain in the mid-sagittal plane. The instructor will assist you in doing this properly. Many of the structures you have already identified will now appear in a new perspective. Some of these structures will appear in the following instruction, but you should be sure to locate all previously mentioned structures that now appear in the midline. Before making the sagittal section, your brain should be cleaned of vessels and pia mater.

The aqueduct of Sylvius or **cerebral aqueduct** runs from the rostral portion of the metencephalon through the mesencephalon to connect the fourth ventricle with the **third ventricle** of the diencephalon. The roof of the aqueduct is made up of the **lamina quadrigemina** with its super-imposed quadrigeminal bodies (colliculi). The portion of the midbrain above the aqueduct is called the **tectum**; the portion below the aqueduct and above the cerebral peduncles is called the **tegmentum**.

In the diencephalon, you will see the **thalamus** with a large central mass, the **massa intermedia**, which extends across the midline. The ventral surface of the thalamus is bounded by the **hypothalamus**. There is no clear separation between thalamus and hypothalamus seen in sagittal section. In very general terms, the hypothalamus integrates visceral-somatic activity and influences the secretion of substances from the pituitary gland. Part of the control of feeding behavior, drinking, sexual behavior, temperature regulation and rage responses, and more is vested in the hypothalamus. The thalamus is a relay station for sensory pathways ascending on the way to the cerebral cortex. The thalamus also has many other diverse functions. Having diffuse reciprocal connections with the cerebral cortex, it plays a part in attention and arousal mechanisms.

Part of the dorsal and anterior limit of the thalamus is formed by the **fornix**, a bundle of fibers arising below the posterior portion of the **corpus callosum** and arching downward to dive below the surface in route to the mammillary bodies. The fornix arises from the band of fibers, the fimbria, which runs along the medial surface of the hippocampus (both fimbria and hippocampus will be seen later). One fornix is connected to the other by transverse fibers called the commissure of the fornix or more commonly, the hippocampal commissure. The portion of the fornix where the two are connected is called the **body of the fornix** while the separate downward arching part is called the **column of the fornix**.

The fornix carries fibers from the hippocampus to the mammillary bodies (via the postcommissural fornix) and septal nuclei (via the precommissural fornix) and fibers from septal nuclei to hippocampus. Near the anterior arch of the fornix, you will note a small round bundle of fibers, the **anterior commissure**, which connects the olfactory bulbs, pyriform area and amygdalae of the two hemispheres.

Completing the anterior border of the diencephalon is the **lamina terminalis**, which forms part of the anterior wall of the third ventricle. At the posterior end of the thalamus, you will again see the pineal body. Immediately anterior to the pineal body is the **habenula**, a thalamic nucleus probably associated with olfacto-somatic correlation. Below the pineal body and anterior to the lamina quadrigemina is the **posterior commissure**, another bundle of fibers connecting the two hemispheres. It forms the border between diencephalon and mesencephalon and is just dorsal to the point where the aqueduct of Sylvius becomes continuous with the third ventricle.

Just below the fornix anterior you should note the **inter-ventricular foramen** or foramen of Monro, which communicates between the **lateral ventricle** (to be studied later) and the **third ventricle**.

Now examine the **corpus callosum**, the large band of fibers that serves to connect the two hemispheres. The sharply bent anterior portion is called the **genu** (meaning knee). As the genu turns under, it narrows into the **rostrum** (meaning beak) of the corpus callosum. The posterior end of the corpus callosum is called the **splenium**. The thin sheet of tissue running between the corpus callosum and fornix is the **septum pellucidum**. It forms the medial wall of the lateral ventricle.

Turning now to the cerebral cortex, it is questionable whether the gyri and sulci to be seen on the medial surface are analogous to those of man. The one primary and constant sulcus of the midline cortex is the **cingulate sulcus**. Between the sulcus and the corpus callosum is the **cingulate gyrus**. The **callosal sulcus** lies between cingulate cortex and corpus callosum. The cortex lying above the cingulate gyrus may be considered together as the **marginal gyrus**. The primary sensory and motor areas recognized on the dorsal surface extend over into the marginal gyrus. The portion of the cingulate gyrus around and behind the splenium of the corpus callosum is called **retrosplenial cortex**. The cortex in front of and below the genu of the corpus callosum is called the **subcallosal gyrus**. Cortex below this is called the **paraolfactory gyrus**. This cortex below the rostrum of the corpus callosum as well as some more laterally placed nuclei is the **septal area**. The subcallosal and paraolfactory gyri are included in the term septal area. You

should recognize that the cingulate gyrus extending around the posterior end of the corpus callosum becomes continuous with the hippocampal gyrus, which you saw on the ventral surface.

As dissection continues, you will be able to better visualize this continuity. The cingulate gyrus together with its continuation as the hippocampal gyrus is often called the **limbic lobe**, as it tends to make a border around the upper end of the brain stem (limbic = border). The term limbic lobe is, however, often used synonymously with limbic system which includes the hippocampus, **parahippocampal gyrus**, amygdala, **septal area**, **cingulate gyrus**, and often the olfactory structures as well; so that it comes to include most of the rhinencephalon as well as additional structures thought to be functionally related. As you can see, the terminology is not generally agreed upon here, and the functional relation of all of these brain areas is via the **fornix**, stria terminalis, stria medullaris, **anterior commissure**, **diagonal band of Broca**, and the medial forebrain bundle.

3. **Cleaning**

Before leaving lab you must properly and carefully store your brain, clean any equipment you used today, and clean your table.

Name _____

Homework

1. Draw a mid-sagittal view of the brain.
 - a) Label the components of the ventricular system
 - b) Indicate the flow of CSF thru the ventricular system, beginning from choroid in lateral ventricles and proceeding to the arachnoid cisterns via 3 foramina.
2. Draw a separate/additional mid-sagittal view of the brain, indicating the key structures

Homework continuation page

Lab 5

The Spinal Cord

Introduction: The spinal cord is the caudal extension of the central nervous system. It's main function is to send and receive nerve impulses to and from the periphery. Sensory and motor information can be integrated in local reflex arcs that do not require central (brain) control. In fact, many complex movements can be observed in animals whose entire cortex has been removed. The spinal cord is divided into 31 segments, each which give rise to one spinal nerve: 8 **cervical**, 12 **thoracic**, 5 **lumbar**, 5 **sacral**, and 1 **coccygeal**. Each spinal nerve innervates a dermatome from which the nerve was derived during embryonic development. When viewed in cross section, the arrangement of spinal neuron cell bodies (**gray matter**) is shaped like the letter H. The surrounding columns of **white matter** are collections of ascending and descending axons. Afferent information travels in to the brain in the **dorsal column** and efferent information travels from the brain in the **ventral column**. The **lateral column** carries both ascending and descending tracts. It is important as you study to associate different cortical pathways with the region they travel in the spinal cord.

***** YOU WILL NEED COLORED PENCILS TODAY*****

The Spinal Cord and Summary of Tracts

1. Ascending / Sensory

(A & B)	Dorsal column / Medial lemniscal	_____
	(Posterior Column)	_____

(E)	Lateral Spinothalamic	_____
-----	-----------------------	-------

(G)	Anterior Spinothalamic	_____
-----	------------------------	-------

Unconscious Spinocerebellar
(C) Posterior
(D) Rostral
(F) Anterior

2. Descending / Motor

a) Lateral (Distal)

- (K) Lateral corticospinal crosses “_____”
Speed, precision, & strength to _____ muscles
Conscious, direct projections to _____
Use fingers independently (pincer grasp)
Skilled rotary movement around limb joints

(Motor cortex to spinal motor neurons)

- (L) Rubrospinal
_____ muscles
Independent arm or limb movement
Input from cortex and cerebellum

(_____ to spinal interneurons)

b) Medial (Proximal: Trunk / Axial)

- (M / M) Reticulospinal (2)
Speed and precision to axial muscles
Enhance motor activity by coordination (walking)

(From reticular formation to spinal cord)
- (N) Vestibulospinal

_____ muscles
_____ & _____
Input from inner ear & cerebellum
(Momentarily inhibited before voluntary movements)

(From vestibular nucleus to spinal cord)
- (O) Tectospinal
Orient to visual stimuli

(From superior colliculi to spinal cord)

Procedure

1. Spinal Cord Organization

- A. Identify (by letter) the following structures on spinal cord slice (A) on the following page.
- B. Color each labeled structure.

Key Structures:

- ___ Dorsal funiculus
- ___ Dorsal horn
- ___ Dorsal intermediate sulcus
- ___ Dorsal lateral sulcus
- ___ Dorsal median septum
- ___ Dorsal median sulcus
- ___ Lateral funiculus
- ___ Lateral horn (intermediate zone)
- ___ Ventral funiculus
- ___ Ventral horn
- ___ Ventral lateral sulcus
- ___ Ventral median fissure

2. Summary of tracts

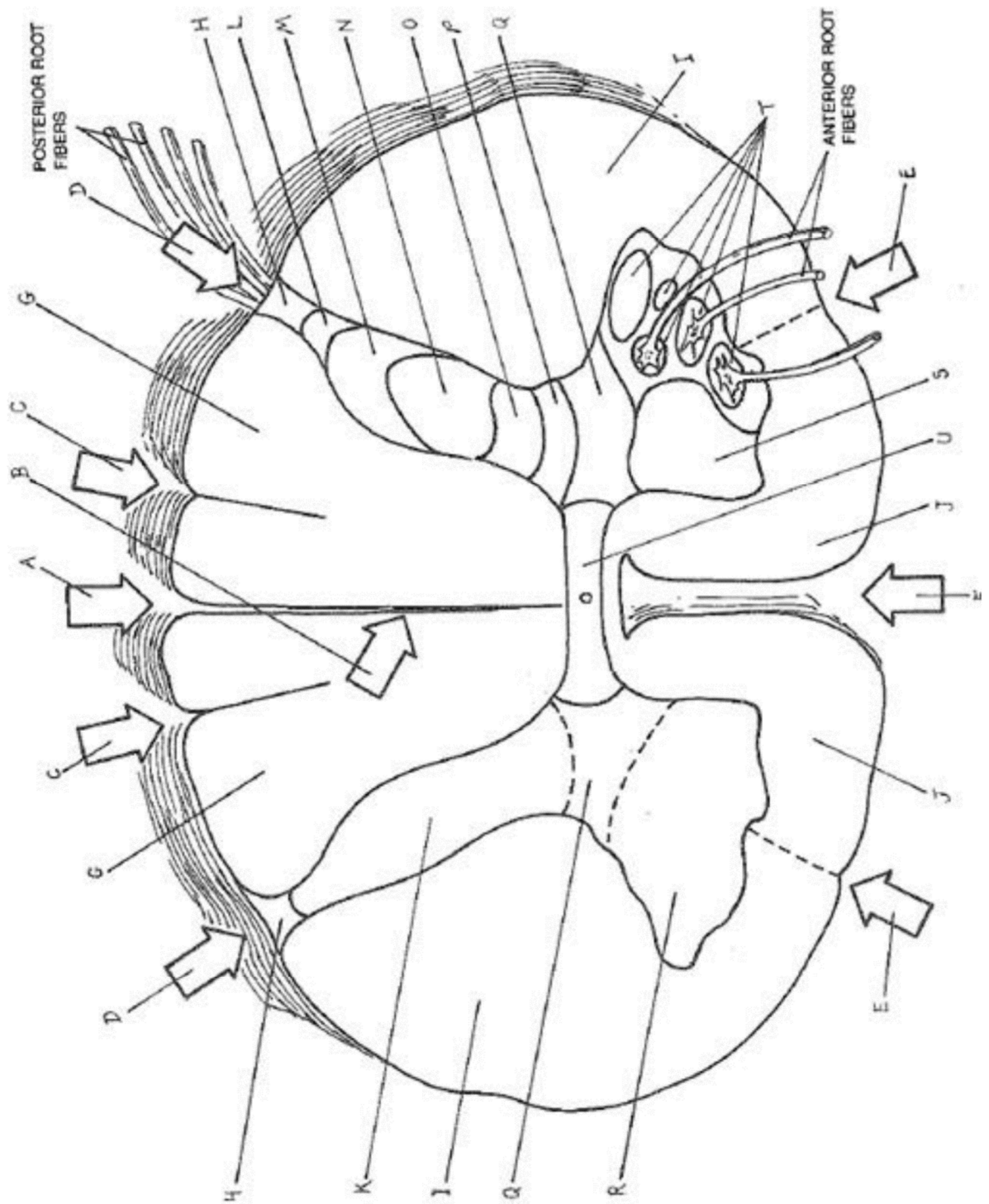
- C. Identify (by letter) the following tracts on spinal cord slice (B) on the next following page.
- D. Color each labeled structure
- E. Identify each tract as either ascending (A) or descending (D)

Key Structures:

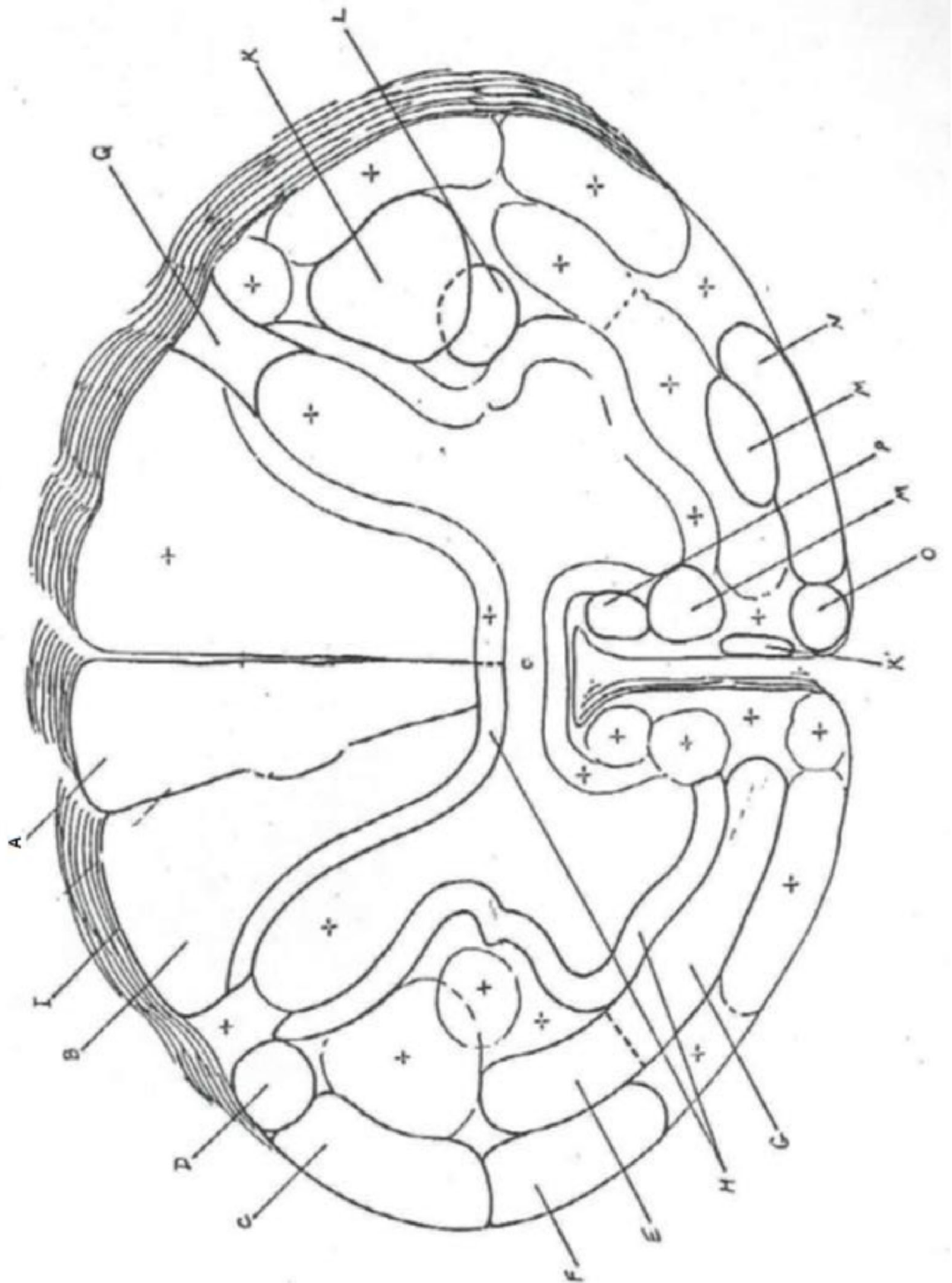
- ___ Anterior corticospinal tract A / D
- ___ Anterior spinocerebellar tract A / D
- ___ Fasciculus cuneatus A / D
- ___ Fasciculus gracilis A / D
- ___ Lateral corticospinal tract A / D
- ___ Lissauer's tract A / D
- ___ Posterior spinocerebellar tract A / D
- ___ Spinothalamic tract (antero-lateral system) A / D
- ___ Reticulospinal tract A / D
- ___ Rubrospinal tract A / D
- ___ Tectospinal tract A / D
- ___ Vestibulospinal tract A / D

Be certain you can identify ALL the structures from the previous labs.

A. Spinal Cord Organization



B. Summary of Spinal Cord Tracts



Name _____

Homework

1. Complete the diagrams of the spinal cord
2. Draw a diagram of the origin and termination of each of the following tracts. Pay attention to laterality (ie, decussation), location in the spinal white matter, and the terminal region of the spinal horn.
 - a) Anterior corticospinal tract
 - b) Lateral corticospinal tract
 - c) Reticulospinal tract
 - d) Rubrospinal tract
 - e) Tectospinal tract
 - f) Vestibulospinal tract

Homework continuation page

Lab 6

Horizontal Structures; The Basal Ganglia

Introduction: In this laboratory, the anatomy and functions of the basal ganglia are discussed. The basal ganglia are a collection of nuclei deep to the white matter of the cerebral cortex, including the structures: the caudate nucleus, putamen, nucleus accumbens, globus pallidus, substantia nigra, and subthalamic nucleus. These areas are important in control of movement and disruptions in the basal ganglia system are seen in disorders of movement, such as Parkinson's and Huntington's disease.

Key Structures:

Alveus	Inferior colliculus
Amygdala	Inferior horn of lateral ventricle
Anterior commissure	Internal capsule
Anterior horn of lateral ventricle	Lateral geniculate nucleus
Anterior limb of internal capsule	Lateral ventricle
Caudate nucleus	Massa intermedia
Cerebral peduncle	Optic tract
Clastrum	Posterior limb of internal capsule
Corona radiata	Pulvinar
Corpus callosum	Putamen
External capsule	Pyramids
Extreme capsule	Septal area
Fimbria	Septum pellucidum
Fornix	Superior colliculus
Globus pallidus	Thalamus
Habenula	
Hippocampus	

Basal Ganglia

The 5 Structural Components (Major Nuclei):

- 1.
- 2.
- 3.
- 4.
- 5.

Terms:

Lenticular or Lentiform Nucleus:

Striatum:

Globus Pallidus (a.k.a.-_____):

Caudate:

Substantia Nigra:

2 types of interactions with B.G. circuitry

The terms give rise to prefixes and suffixes that describe fibers coming from / going to parts of the basal ganglia.

Striatum: _____ - and - _____. Striatopallidal fibers go from _____ to _____. Corticostriate fibers go from _____ to _____, _____, or _____.

Globus Pallidus: Pallidothalamic fibers go from _____ to _____.

Pathways:**Parallel Loops - General Pathway**

Projections from cortex → _____ & _____ = Principal input

Excitatory (_____) connections = corticostriate inputs & projections to cortex from thalamus

Inhibitory (_____) connections = GPi and SNr outputs

Direct:

Indirect:

Disorders:

Huntington's Disease:

Parkinson's Disease:

Procedure

To view the structures within your sheep brain, you will need to make horizontal cuts through your sheep brain. These cuts may be difficult to make, carefully follow the directions your lab instructor gives as well as the directions given in the dissection guide. Once again, do not cut back and forth like a saw; this will damage structures. If you have any questions or are uncertain about this process, alert your lab instructor at once, it is imperative that these be good cuts to ensure that you can identify all the necessary structures.

Briefly, you will make 4 cuts through your sheep brain. Angle your blade parallel to the callosal sulcus, rather than parallel with the table. The first cut exposes the corona radiata; the second is through the top of the lateral ventricle; the third cut is through the dorsal thalamus; and the final cut is through the anterior commissure. Your lab instructor will give you more specific instructions before you begin.

- 1. Follow Instructions**

Carefully read and understand the instructions on the cuts you are about to make.

- 2. Dissection Guide**

Deep Structures of the Brain – Horizontal sections

A. Horizontal section opening the lateral ventricle.

Tear away the **septum pellucidum** and examine the lateral ventricle. Note that the corpus callosum forms the roof of a large part of the cavity. Identify the head of the **caudate nucleus** in the anterior floor of the ventricle. In the medial and posterior floor of the ventricle, identify the **hippocampus** and **fimbria** of the hippocampus running along its lateral and anterior edge. Note the position of these structures now. You will, however, be able to see them more clearly when you have opened the roof of the ventricle by horizontal section.

With a knife, make successive, thin horizontal sections through the cerebral cortex until you open the roof of the ventricle. Proceed cautiously, so as not to destroy structures before you examine them. As you progress with these sections, note the white matter as it projects into the gyri of the cerebral cortex. This white matter, the **corona radiata**, is composed of sensory fibers ascending from the thalamus and motor fibers descending to lower structures. As you proceed with horizontal sections, you will be able to follow these motor fibers downward into the **internal capsule**, **cerebral peduncles**, and **pyramids** of the medulla. The radiations of the corpus callosum are also mingled with this white matter.

You should continue this section until you have opened the roof of the ventricle and cut through the dorsal part of the corpus callosum on a plane parallel to the callosum. Now again examine the ventricle. The head of the caudate in the anterior floor of the ventricle plunges downward in the anterior horn of the lateral ventricle until it becomes continuous with the anterior perforated substance. The narrowing, backward, and downward direction portion of the caudate (the **tail of the caudate**) loops around the **thalamus** to end at the **amygdala** in the parahippocampal gyrus. Running with the caudate along its medial edge at its junction with the thalamus is the stria terminalis, which carries fibers from amygdala to septal area.

In the posterior floor of the ventricle, note once again the hippocampus and fimbria. The **hippocampus** is covered by a thin layer of white matter, the **alveus**, which becomes continuous with the **fimbria**, which is in turn continuous with the **fornix**. If you gently scrape a small area of the alveus with a scalpel, you can expose the underlying gray matter of the hippocampus, which arches downward around the thalamus in a manner similar to the caudate and ends in the **hippocampal gyrus** posterior and lateral to the amygdala. ***By making careful, thin frontal sections beginning at the occipital pole, you can open the back of the lateral ventricle and see the arch of the hippocampus as it turns downward in the **inferior horn of the lateral ventricle**. Keep in mind these overall descriptions of the caudate and hippocampus, and their anatomical structure will become clear to you as you continue horizontal section.***

When properly made, these sections should cut through the dorsal portion of the corpus callosum on a plane parallel to the callosum. Your sketch should include and identify all of the structures seen in the ventricle, the internal capsule, corona radiata, corpus callosum, lateral fissure, cerebral cortex, and the ventricle. On this and all other sketches, you must indicate anterior-posterior and lateral-medial. All structures and landmarks revealed by a section should be identified whether or not specified in the instructions and whether or not the structure has appeared and been identified in another sketch. Do not fail to include such overall landmarks as the cerebral cortex, lateral fissure, etc., when they appear.

B. Horizontal section through the dorsal thalamus.

Continue the horizontal sections until you reach the dorsal surface of the thalamus. This section, when properly made, should be parallel to the plane of the corpus callosum and just pass through the dorsal surface of the massa intermedia. It should pass above the pineal body posteriorly and just below the genu of the corpus callosum anteriorly.

Anteriorly, you will see the **head of the caudate nucleus** in the **anterior horn of the lateral ventricle**. Proceeding laterally from the head of the caudate, the white matter is the **internal capsule**, which extends posteriorly in an oblique L-shape. Gray matter lateral to the anterior half of the capsule is the **putamen** bounded laterally by a thin, white stripe, the **external capsule**. Lateral to this capsule is a band of gray, the **claustrum**. Medially across the ventricle from the head of the caudate is the **septal area** bounded posteriorly by the cut edge of the **fornix**. Posterior to the **head of the caudate** and medial to the internal capsule is the central gray mass of the thalamus with the medially placed **massa intermedia** and **habenula**, the posteriorly placed **pulvinar** and lateral to the pulvinar on the border of the thalamus, the gray of the **lateral geniculate nucleus** should appear surrounded by the white of the **optic tract**. Lateral to the geniculate, mingled with the fibers of the posterior part of the internal capsule, is the **tail of the caudate** and the associated stria terminalis. The stria may not appear as a bundle distinct from the internal capsule at this level. Posterior to the geniculate is the **hippocampus** and **fimbria** running in the **posterior** and **inferior horns of the lateral ventricle**. Posteriorly, of course, the cerebellum will appear.

C. Horizontal section through anterior commissure.

Continue the horizontal sections until you reach a level through the anterior commissure and superior colliculus posteriorly. Most of the structures that appeared in the last sketch will still be seen with a slightly different perspective. The **globus pallidus** should now appear between the **putamen** and **internal capsule**. Note how the **anterior commissure** swings forward in a U-shape serving to interconnect the two olfactory bulbs. Posteriorly, the tail of the caudate will appear more distinctly than in the previous section.

D. Horizontal sections to the optic chiasm.

Continue these horizontal sections downward to clarify as many anatomical relationships as possible. Be constantly alert to the fact that you are revealing the internal position of many structures that you have already identified on the surface of the brain. At the level of the inferior colliculus the medial geniculate will appear and the associated brachium of the inferior colliculus. Note the relationships of the **hippocampus**, **amygdala**, and **parahippocampal gyrus**. The **cerebral peduncles** will be revealed when your horizontal sections reach a level just above the optic chiasm. The peduncles are, of course, the motor fibers you have been following throughout your horizontal sections; first as the **corona radiata**, then as the **internal capsule**, and they continue downward into the medulla as the **pyramids**.

3. Cleaning

Before leaving lab you must properly and carefully store your brain, clean any equipment you used today, and clean your table.

Name _____

Homework

Draw representative slices of sheep brain from each of the following levels. Label all structures present at each level.

1. Horizontal section through the dorsal thalamus.
2. Horizontal section through anterior commissure.
3. Horizontal sections through the amygdala and hippocampus.

Homework continuation page

Lab 7

The Cerebellum

Introduction: In this laboratory, the anatomy and functions of the cerebellum will be examined. Cerebellum literally means “little brain”, and it contains as many neurons as the rest of the central nervous system. The cerebellum is involved in the coordination of movement, motor learning, and motor reflexes. The cerebellum works in such a way as to compare what you think you are going to do (from motor cortex) with what is actually going on down in your limbs and muscles (such as proprioceptive information feeding back). It can then help to make any necessary corrections for problems that arise. There is evidence that the cerebellum is also partly responsible for motor learning, such as riding a bicycle or playing a piano. After this lab, you should be able to identify the following structures and terms.

***** YOU WILL NEED COLORED PENCILS TODAY*****

Key Structures

4th Ventricle	Ansiform
Anterior lobe	Cerebellar hemispheres
Anterior medullary vellum	Dorsal intermediate sulcus
Arbor vitae	Dorsal median fissure
Central canal	Fasciculus cuneatus
Central lobule (Centralis)	Fasciculus gracilis
Cisterna magna	Flocculonodular lobe (flocculus + nodulus)
Culmen	Flocculus
Foramen of Magendie	Inferior cerebellar peduncle
Lingula	Middle cerebellar peduncle
Middle lobe	Paraflocculi
Nodule	Paramedian lobules
Obex	Superior cerebellar peduncle
Posterior lobe	Vermis cerebelli
Primary fissure	
Pyramis	
Simplex lobule	
Tuber vermis	
Uvula	

The Cerebellum

The divisions and their function:

Flocculonodular Lobe = vestibulocerebellum

Comprised of flocculus & nodule

Functions

Vermis & Paravermis = spinocerebellum

Comprised of midline vermis & paravermis

Functions

Lateral hemispheres = cerebrocerebellum

Comprised of lateral hemispheres

Functions

Cerebellar Peduncles

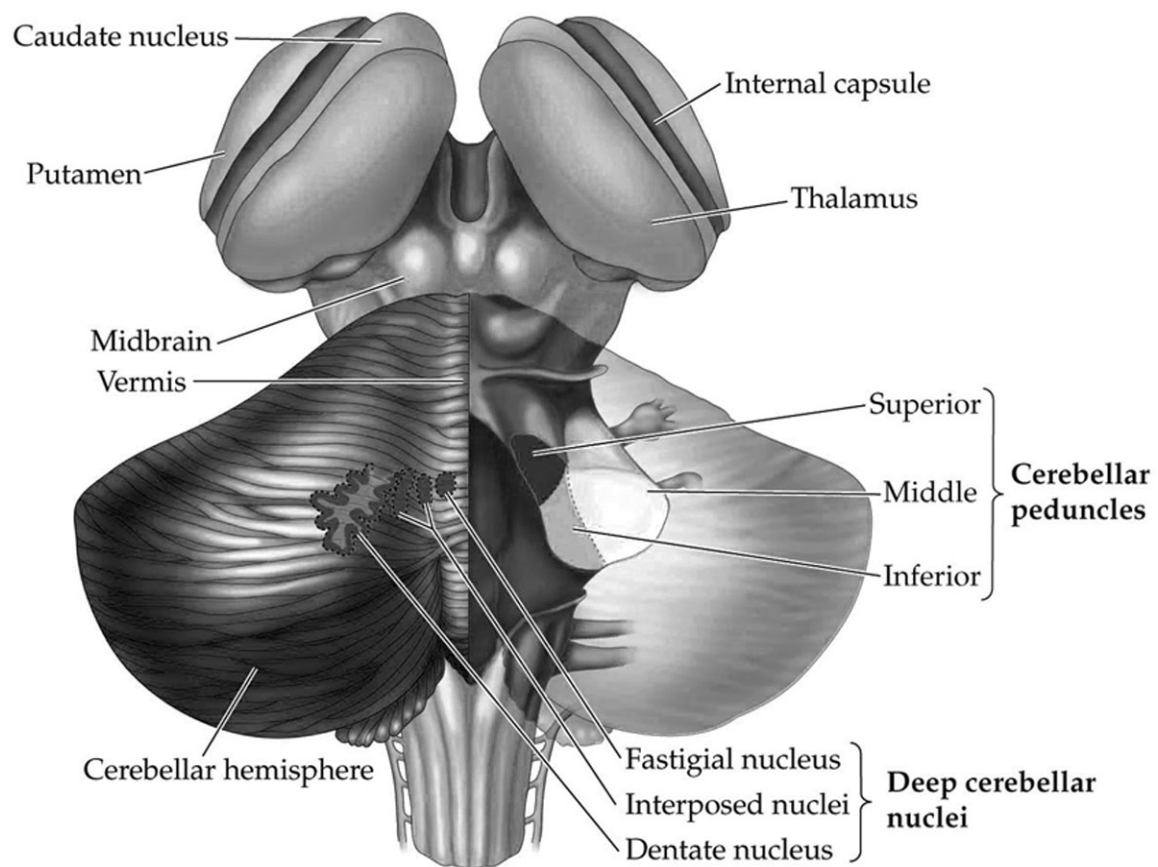
Inferior cerebellar peduncle

Middle cerebellar peduncle

Superior cerebellar peduncle

Dysfunction of the Cerebellum

Vermis Cerebelli (medial)	Cerebellar Hemispheres (lateral)
<u>Anterior Lobe</u>	
(Primary Fissure)	
<u>Middle Lobe</u>	
<u>Posterior Lobe</u>	
Posterior Median Lobule	
<u>Flocculonodular Lobe</u>	



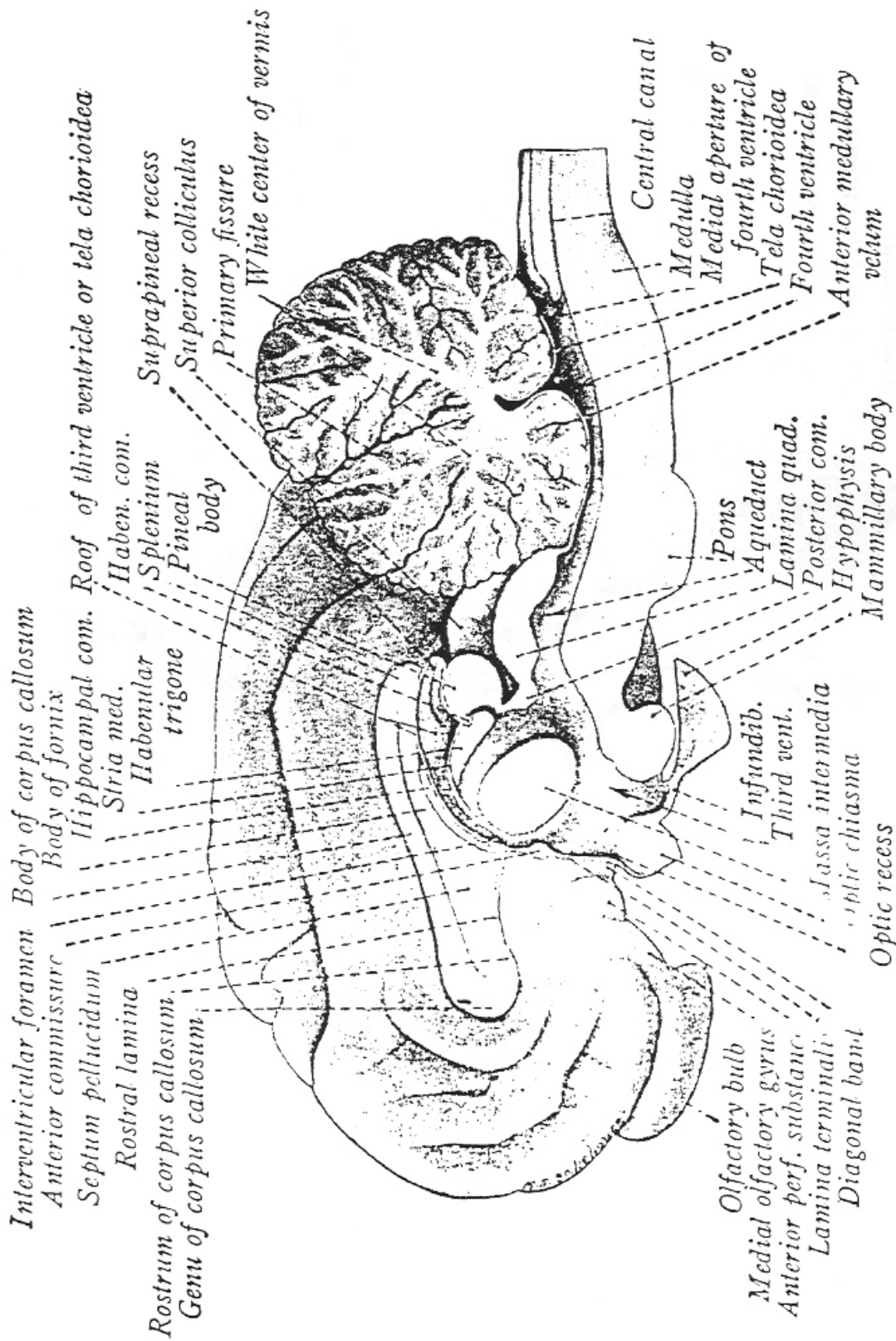


FIGURE 419. Medial sagittal section of the sheep's brain.

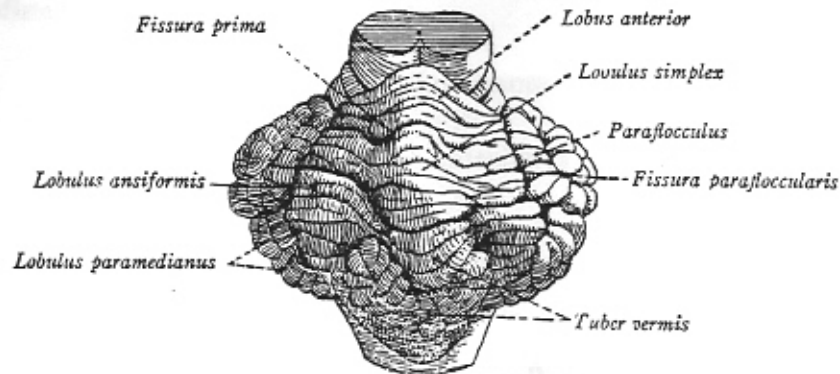


FIGURE 424. Cerebellum of the sheep, dorsorostral view.

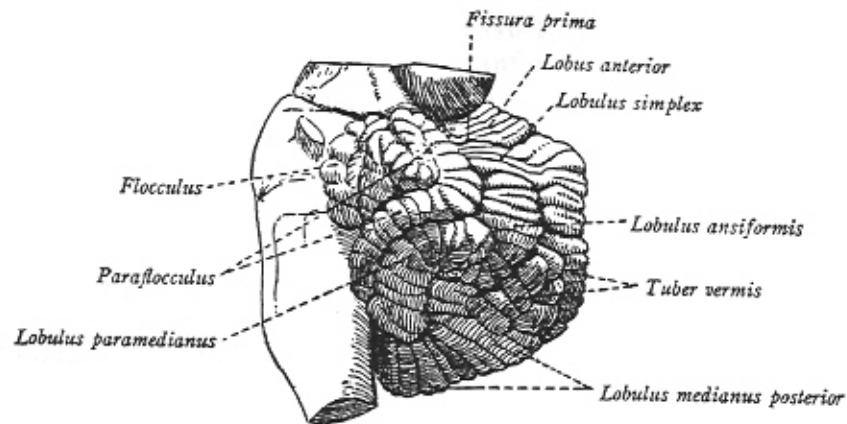


FIGURE 425. Cerebellum of the sheep, lateral view.

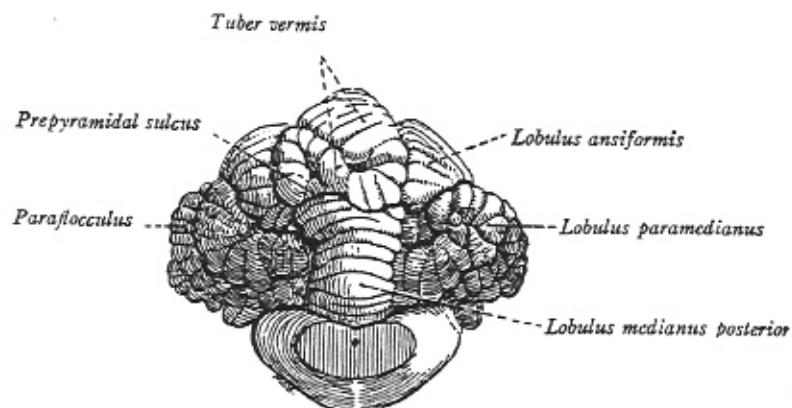


FIGURE 426. Cerebellum of the sheep, caudal view.

Procedure

Be able to identify the previously listed cerebellar structures on the sheep brain specimens.

1. Dissection Guide

External anatomy

F. The cerebellum, fourth ventricle, and dorsal view of the brain stem.

Remove the cerebellum by taking horizontal sections of it. As you make these sections, note how the white central core projects into the lobules and folia. As you continue these horizontal sections, and by reflecting the cerebellum in various directions as you progress, you should note once again the **middle cerebellar peduncle** (brachium pontis) as it enters the cerebellum as a continuation of the transverse fibers of the pons carrying primarily pontocerebellar fibers. Entering the cerebellum medially to the brachium pontis and extending upward from the cerebellum to end just beneath the inferior colliculus is the **superior cerebellar peduncle** (brachium conjunctivum), which connects midbrain structures with the cerebellum. The **inferior cerebellar peduncle** (restiform body) enters the cerebellum caudally carrying fibers to and from the medulla oblongata. All afferent and efferent fibers of the cerebellum caudally carrying fibers of the cerebellum must pass through one of these peduncles. You can now also identify the dorsal cochlear nucleus, a compact crescent-shaped mass of gray matter encircling the restiform body at the point where the latter turns dorsal-ward to enter the cerebellum. The **eighth cranial nerve** will be seen associated with this nucleus at its lateral extent. The dorsal cochlear nucleus (along with the ventral cochlear nucleus which is not visible from the surface) is the first place the eight nerve fibers synapse on their route from the cochlea to higher brain centers. The acoustic area is seen in the floor of the fourth ventricle medial to the dorsal cochlear nucleus. It overlays the vestibular nuclei, which receive fibers from the labyrinth of the inner ear via the vestibular part of the eighth nerve.

G. Myelencephalon.

The dorsal surface of the medulla is largely concealed by the **cerebellum**. By lifting the cerebellum away from the medulla, you can identify the **fasciculus gracilis** (and its rostral extension called the **clava**) and the more laterally placed **fasciculus cuneatus**, longitudinal fiber tracts moving from spinal cord to thalamus. Lateral to cuneatus you will find the **tuberculum cinereum** marking the spinal nucleus and tract of the fifth cranial nerve (trigeminal). The intersection of the lateral walls of the fourth ventricle is called the **obex**. Also, identify the **dorsal** (or posterior) **median fissure**, separating the gracile fasciculi in the midline. It may be difficult to visualize these dorsal myelencephalon structures without disrupting the **cisterna magna**. If so, delay your study of them until you have seen the metencephalic structures indicated below.

H. Metencephalon.

The cerebellum, like the cerebral cortex, is thrown up in convolutions called folia. The cerebellum is composed of an unpaired median portion, the **vermis cerebelli**, and two lateral masses, the **cerebellar hemispheres**. Four lobes comprise the cerebellum. Starting at the abutment of the cerebellum and superior colliculi, locate the **anterior lobe**.

The **primary fissure** separates the anterior lobe from the **middle lobe**, which is made up of the anteriorly placed **simplex lobule**, the medially placed **tuber vermis**, and a lateral portion consisting of the paired **ansiform** and **paramedian lobules**. The posterior median lobule and the paired **paraflocculi** make up the **posterior lobe**. The **flocculonodular lobe** consists of the paired **flocculi**, which are placed far laterally adjoining the pons, and the **nodule**, which can be seen by lifting the cerebellum away from the medulla and viewing down the medulla.

Mid-sagittal Section

Starting at the caudal end of the myelencephalon, note the **central canal**, which extends caudally into the spinal cord and rostrally into the **fourth ventricle**. You should now be better able to visualize the **foramen of Magendie**, communicating between the fourth ventricle and **cisterna magna**. In the metencephalon, note the fourth ventricle lying beneath the cerebellum. The posterior roof of the ventricle is referred to as the tela choroididea, an infolding of the pia mater from which the choroids plexus arises. The anterior roof of the ventricle is called the **anterior medullary velum**. Examining the median view of the cerebellum, you will see the white mater of the core as it projects tree-like, hence the name **arbor vitae**, into the folia of the cerebellum. You can now more clearly see the deep fissures of the cerebellum, which divides it into two lobes. Starting at the anterior medullary velum there is a small lobule, the **lingula**, followed by the **central lobule**, and then several small lobules up to the **primary fissure** called collectively the **culmen**. All these lobules make up the **anterior lobe**. In the **middle lobe**, the anterior lobule is **declive** (seen on the external surface as part of the **simplex lobule**) and the posterior is the **tuber vermis**. The **posterior lobe** includes the **pyramis** adjoining the tuber and the posterior **uvula**. The last lobule is the **nodule**, the median part of the **flocculonodular lobe**.

2. Cleaning

Before leaving lab you must properly and carefully store your brain, clean any equipment you used today, and clean your table.

Name _____

Homework

Draw the cerebellum and label all key structures and peduncles in each of the following aspects:

1. Superior view
2. A median section through the vermis
3. Lateral view

Homework continuation page

Lab 8

Coronal Structures

Introduction: Upon the completion of the examination of these areas you and your lab partner will make coronal sections of your sheep brains and follow the dissection guide in your investigation of these structures. You should be able to identify many of these key structures in coronal slices from several different rostro-caudal levels.

Key Structures

3rd ventricle	Hypothalamus
Amygdala	Internal capsule
Anterior commissure	Lateral fissure
Anterior perforated substance	Lateral geniculate nucleus
Body of the corpus callosum	Lateral ventricle
Callosal sulcus	Lentiform nucleus (putamen + globus pallidus)
Caudate	Mammillary body
Cerebral peduncles	Mammillothalamic tract
Cingulate gyrus	Medial geniculate nucleus
Cingulate sulcus	Optic chiasm
Clastrum	Parahippocampal gyrus
Corona radiata	Pulvinar
Corpus callosum	Putamen
External capsule	Rhinal fissure
Fimbria	Rostrum of the corpus callosum
Fornix	Septal area
Genu of the corpus callosum	Septum pellucidum
Globus pallidus	Thalamus
Habenula	
Habenulo-interpeduncular tract	
Hippocampus	

Procedure

To view the internal structures of your sheep brain, you will need to make coronal cuts through your sheep brain. Using a knife provided by your lab instructor, you will make several cuts, applying gentle downward pressure. Do not cut back and forth like a saw; this will damage the structures. If you have any questions or are uncertain about this process, alert your lab instructor at once, it is imperative that these be good cuts to ensure that you can identify all the necessary structures. There may be structures discussed in the dissection guide and on the list that will not be in the sheep brain atlas, you are still responsible for ALL structures.

1. Follow instructions

Carefully follow the instructions your from your lab instructor and the dissection guide.

2. Dissection Guide

In cutting and visualizing frontal sections, recall how your horizontal sections looked at various levels. This will help you identify structures seen frontally and will allow you to develop a three dimensional view of the brain. Make use of available horizontal sections, brain diagrams, and your own sketches.

A. Frontal section through the anterior perforated substance.

Start your frontal sections from the frontal pole and progress backward slowly and carefully. As you progress, note how the radiations of the **corpus callosum** blend into the **corona radiata** and how the **genu of the corpus callosum** forms the medial floor of the **lateral ventricle**. Continue the sections until you reach a level through the **anterior perforated substance** just anterior to the **optic chiasm** and passing through the **rostrum of the corpus callosum** just posterior to the **genu**, so that the rostrum is separated from the **body** of the corpus callosum.

When you reach this level, note the **head of the caudate nucleus** in the **anterior horn of the lateral ventricle**. Moving laterally and slightly ventral-ward from the caudate, you will encounter the somewhat scattered fibers of the **anterior limb of the internal capsule**. The internal capsule separates the head of the caudate from the lentiform nucleus. (Only the **putamen** of the lentiform nucleus is clearly visible at this level. As you dissect backward, the internal capsule will become a more distinct bundle and will clearly separate the caudate from the lentiform nucleus and the **globus pallidus** will appear ventro-medially to the putamen.) This section shows clearly the source of the collective name of the caudate and lentiform (corpus striatum = striped body) as the internal capsule stripes the nuclei.

Lateral to the putamen, the distinct thin, white stripe is the **external capsule** and the more or less distinct gray stripe lateral to the capsule is the **claustrum**. The large solid mass of white matter located centrally is the **corona radiata** and the radiations of the corpus callosum project into it.

Medially on this section, you should note the **rostrum of the corpus callosum** and the **septal area** beneath it. If the **septum pellucidum** is still intact, you can note its

continuity with the septal area. Above the ventricle, you will find the **body of the corpus callosum** and the **cingulate gyrus** and **sulcus**. You should also note on this section the **lateral olfactory stria**, **anterior perforated substance**, **lateral fissure**, and **rhinal fissure**.

B. Frontal section through the anterior commissure.

Continue backward until you reach a level passing through the **optic chiasm** and just anterior to the **anterior commissure**. In addition to the structures seen on the previous section, this section will clearly reveal the **globus pallidus** separated from the **putamen** by a distinct white stripe, the **external medullary lamina**. Medially, the **preoptic area of the hypothalamus** appears just above the optic chiasm. Note the distinct bundle formed by the forward directed fibers of the U-shape of the **anterior commissure**. Ventrally, the **diagonal band of Broca** will have appeared along with the **parahippocampal gyrus** laterally. This section should show the depth of the **lateral fissure** and should have just opened the **third ventricle** above the optic chiasm.

C. Continuation of frontal sections through the thalamus.

As you continue these frontal sections backward, you will begin to cut through the **fornix**. As your section passes just behind the anterior commissure, you will be cutting the fornix twice, dorsally and ventrally. It will appear ventrally as a compact, round bundle below the middle of the section. As you progress backward, the round bundle of the fornix will appear lower and lower until it reaches the **mammillary body**. Your sections will soon pass behind the **lentiform nucleus** and the continuity of the **cerebral peduncles** with the **internal capsule** will become apparent. The gray mass of the thalamus medial to the **internal capsule** will become more prominent, as will the **hypothalamus**.

D. Frontal section through the mammillary body.

Continue frontal sections until you reach a level just through the anterior part of the mammillary body. (You should have followed the fornix so that you can now recognize it in the mammillary body.) Dorsal to the fornix you should be able to identify the **mammillothalamic tract** as a similarly shaped bundle of fibers. The **cerebral peduncles** will appear clearly on this section along with the upward extension as the **internal capsule**. Note the optic tract lateral to the peduncles. At the junction of the **parahippocampal gyrus** with the **internal capsule** the stria terminalis will appear as a compact bundle of fibers. You should also find the dorsal continuation of the stria terminalis just below the **tail of the caudate nucleus** at its junction with the **thalamus**. You can trace this bundle around the thalamus by carefully separating the hippocampal gyrus from the brain stem. Be careful not to break the stem away from the cerebral hemispheres. Also seen by pulling up the **parahippocampal gyrus** is the **amygdala**. This section also cuts the **fimbria** (or fornix, depending upon the exact level and angle of your cut) and **hippocampal commissure** dorsally. The **thalamus** appears at about its greatest extent in this section and you will note the rough demarcation of the thalamus into various nuclei by some more or less distinct white bands of fibers. Medially, on the dorsal surface of the thalamus is the stria medullaris thalami, one of the pathways of the limbic system serving to connect the septal area, amygdala, parts of the hypothalamus, and other structures with the habenula. The **hypothalamus**, including the **mammillary body**, appears ventrally on this section.

E. Frontal section through the habenula.

As you progress now you may (depending on your angle of cut) sever the remaining connections holding the brain stem to the cerebral hemispheres and the brain stem will fall away from the hemispheres. You should try to maintain the integrity of the whole brain until you complete this section.

Proceed with frontal sections until you reach the anterior part of the **habenula** and expose both the **dorsal** and **ventral hippocampus**. The ventral hippocampus can be easily identified surrounded by the fimbria within the hippocampal gyrus. The dorsal hippocampus appears above the thalamus. On the lateral edge of the thalamus the crescent-shaped (or knee-shaped; geniculate = little knee) **lateral geniculate nucleus** will appear. Medial to it and separated from it by a thin band of fibers is the **medial geniculate nucleus**. On the dorsomedial surface, this section will cut through the **habenula** and a distinct fiber tract coursing downward from it, the **habenulo-interpeduncular tract**, carrying fibers from the habenula to the interpeduncular nucleus of the midbrain. This is one of the pathways by which limbic system influence, which has reached the habenula via the stria medullaris thalami, may reach the mesencephalic reticular formation. The **pulvinar** comprises the bulk of the posterior part of the thalamus at this level.

By making careful, thin frontal sections beginning at the occipital pole, you can open the back of the **lateral ventricle** and see the arch of the **hippocampus** as it turns downward in the **inferior horn of the lateral ventricle**.

3. Cleaning

Before leaving lab you must properly and carefully store your brain, clean any equipment you used today, and clean your table.

Name _____

Homework

Draw representative slices of sheep brain from each of the following levels. Label all structures present at each level.

1. Frontal section through the anterior perforated substance.
2. Frontal section through the anterior commissure.
3. Frontal section through the mammillary body.
4. Frontal section through the habenula.

Homework continuation page

Lab 9

Autonomic Nervous System

I. Autonomic Nervous System

Introduction: The autonomic nervous system maintains homeostasis in the periphery by innervating smooth muscle, cardiac muscle, and glands. It is responsible for the important task of keeping the chemistry, pH, nutrient support, and blood flow of the body regulated within a very narrow range to ensure survival. Whereas skeletal muscle is under voluntary control, the activity of the autonomic nervous system usually occurs subconsciously. There are three divisions of the autonomic nervous system: the sympathetic, parasympathetic, and enteric systems.

A. Overview

Homeostasis

Autonomic or Visceral Nervous System

3 Divisions:

3 Neuron Systems:

Upper (hypothalamus to spinal cord level) thru lateral horn

Preganglionic

Postganglionic

	Sympathetic	Parasympathetic	Enteric
Nickname	_____	_____	_____
Preganglionic NT	_____	_____	_____
Postganglionic NT	_____	_____	_____
Receptors	_____	_____	_____
Effect	_____	_____	_____
Ganglia	_____	_____	_____
Preganglionic fiber length	_____	_____	_____
Postganglionic fiber length	_____	_____	_____
Fiber ratio (pre/post)	_____	_____	_____
Main organ	_____	_____	_____

B. Receptor Subtypes / Functions

Sympathetic

Alpha –

Beta 1 –

Beta 2 –

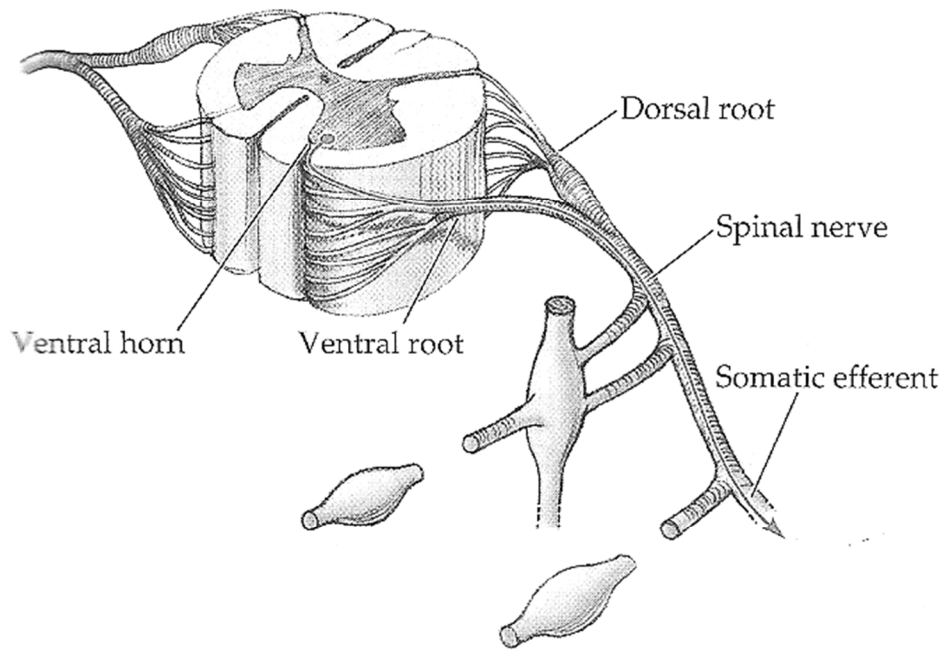
(Dopamine) –

Parasympathetic

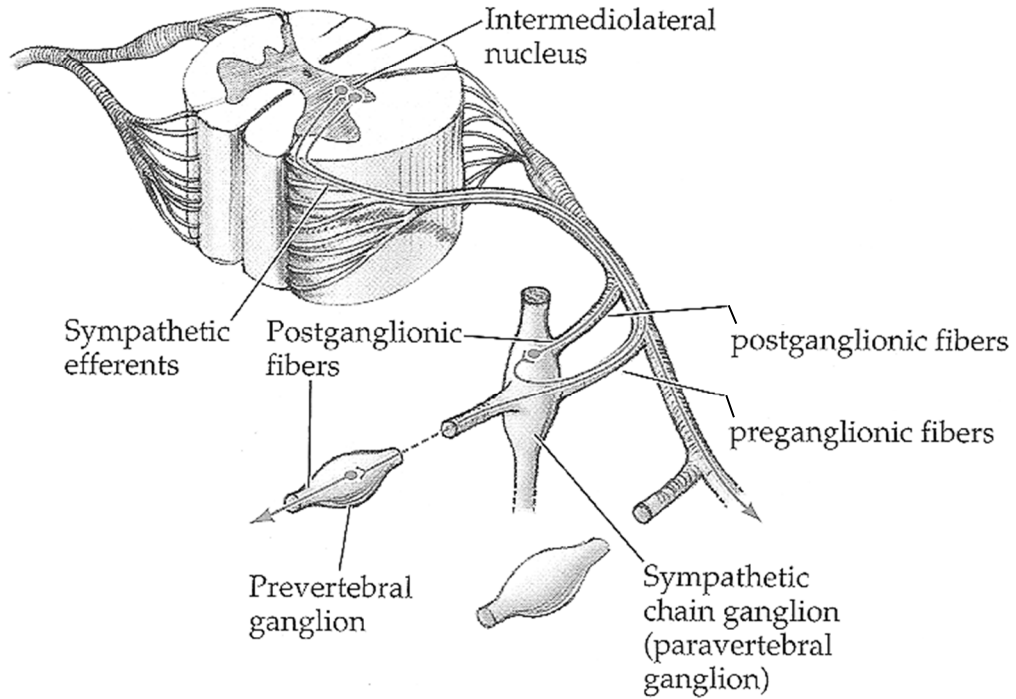
Nicotinic –

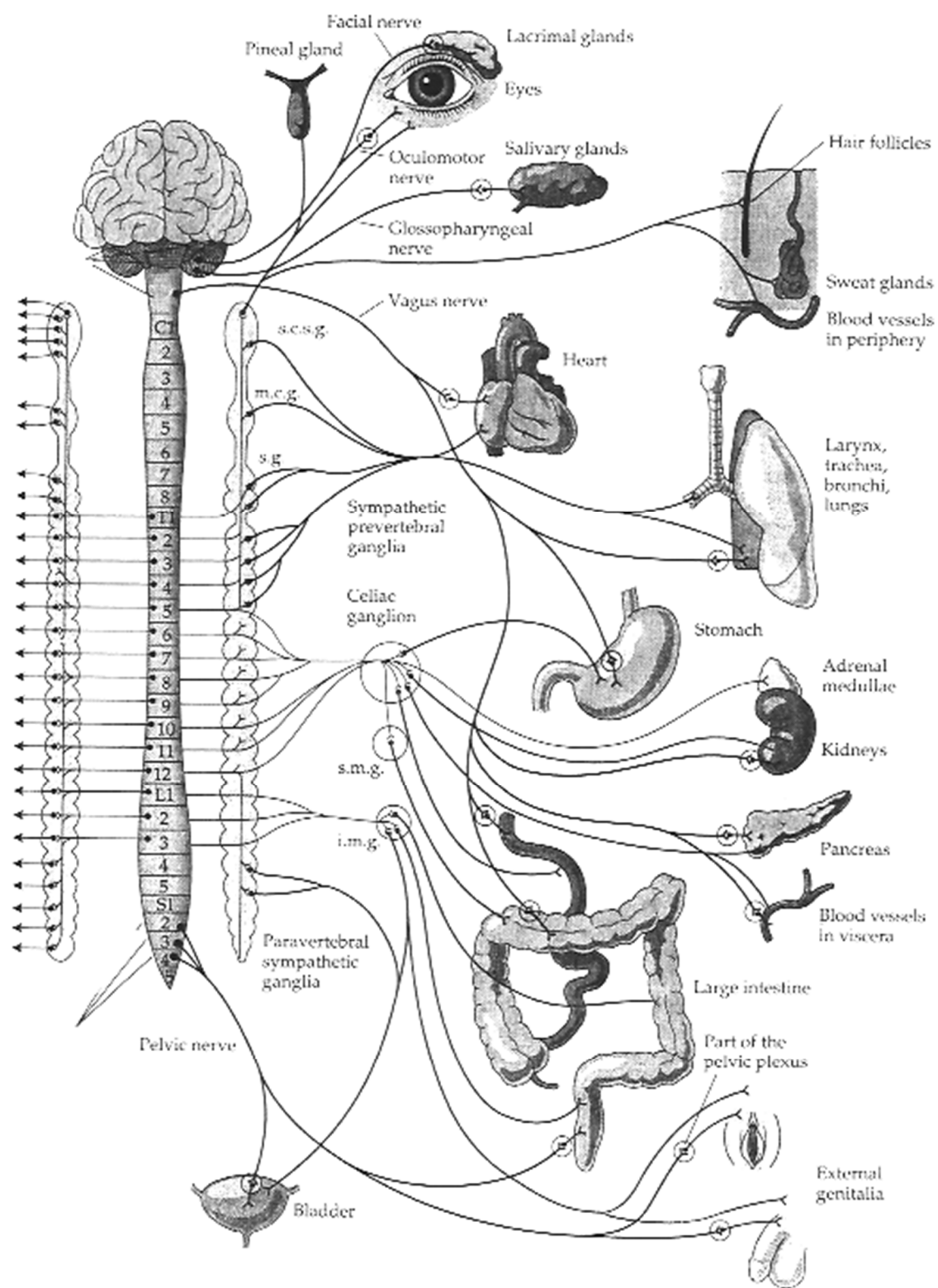
Muscarinic –

(A) Somatic efferent



(B) Sympathetic efferent





Lab 10

Limbic System & Hippocampus

Introduction: The limbic system is a system of interconnected structures in the forebrain that are closely associated with emotional response. It is located on the medial aspect of each cerebral hemisphere and diencephalon, and it consists of the septal nuclei, cingulate gyrus, parahippocampal gyrus, the hippocampus, the amygdala, the hypothalamus, and the anterior nucleus of the thalamus. The word *limbus* means “ring;” thus, it makes sense that the limbic structures encircle the upper part of the brain stem.

Emotional responses are not the only things the limbic system is responsible for. The hippocampus and the amygdala are both important in the formation of memories. The hippocampus is important for laying down new memories into long-term memory storage. The amygdala is important for broader aspects of memory, like if a situation was “good” or “bad.”

The hippocampus is a curved structure in the limbic system. It consists of three parts: the dentate gyrus, the hippocampus proper (or cornu ammonis), and the subiculum. The hippocampus is involved in learning and the formation of long-term memory. Removal of the medial temporal lobes causes anterograde amnesia, where the patient can no longer remember new facts. Explicit memory for new information is lost, but implicit memory is unaffected. The person can still learn how to do new things.

Pierre Paul Broca’s *Le Grande Lobe Limbique* (1878)

Limbus means: _____

Limbic function: _____

James Papez’s Circuit (1937)

Major Players

A. Cingulate cortex:

B. Hippocampus:

Three distinct zones: _____, _____ & _____.

Input from:

1. the parahippocampal gyrus
2. the amygdala
3. reciprocal connections within the hippocampal formation

Projects (output) to:

1. parahippocampal gyrus (which has extensive connections to the cortical sensory areas)
2. fornix
3. dentate only connects to other parts of the hippocampal formation

Functions include:

1. new declarative memories
2. spatial memory; “place cells”

Dysfunction

1. Bilateral temporal lobectomy
2. Korsakoff’s psychosis

C. Amygdala:

Receives projections from:

Projects to:

Functions include:

D. Septal area:

E. Hypothalamus :

F. An expanded concept of limbic system

A. Prefrontal cortex (PFC)

B. Mesotelencephalic reward pathway

Ventral tegmental area (VTA)

Nucleus accumbens (or ventral striatum)

Procedure

Identify the limbic structures in the coronal sections of your sheep brain, referring to the following sections of the dissection guide: *III. External features of the brain (D. & E.); IV. Mid-sagittal section of the brain; V. Deep brain structures (B. – L.)*

- 1. Follow instructions**

Carefully follow the instructions your from your lab instructor and the dissection guide.

- 2. Review the limbic structures** already identified on the external surfaces (**Lab 3**), the mid-sagittal section (**Lab 4**), and the deep structures seen in horizontal (**Lab 6**) and coronal sections (**Lab 8**).

- 3. Dissection of the hippocampus**

Remember never to cut back and forth like a saw; this will damage the structures. If you have any questions or are uncertain about this process, alert your lab instructor at once, it is imperative that these be good cuts to ensure that you can identify all the necessary structures.

At this point, the cerebral hemispheres will have separated from the brain stem. You can now study each separately. In the hemisphere, you can roll the **hippocampus** out of the **posterior** and **inferior horn of the lateral ventricle** and clearly visualize its full extent. With the brain stem thus removed, you can also clearly see the continuity of the **hippocampal gyrus** with the **cingulate gyrus** around the **splenium of the corpus callosum**. This view with the brain stem removed should suggest to you how the name limbic lobe came to be applied to the cingulate and hippocampal gyri as they form a border around the brain stem (limbus = border). These structures and structures anatomically and functionally related to them are thus called the limbic system.

In the brain stem, the lateral and medial geniculate will become more prominent as you continue backward. At a point slightly behind the frontal section through the habenula, you may be able to see the full extent of the **habenulointerpeduncular tract** as it courses between the **habenula** and the interpeduncular nucleus. Slightly further back you should note the crossing fibers of the **posterior commissure**. When your dissection reaches the **superior colliculus**, its layered fiber core will appear along with the **commissure of the superior colliculus**. Ventrally, the **brachium of the inferior colliculus** will appear clearly. The **aqueduct of Sylvius** is surrounded by gray matter, the **periaqueductal gray**.

Within the core of this level of the brain stem are many fibers of passage, which include the fibers you have seen as the **cerebral peduncles**. These will dive through the **pons** and appear on the other side as the **pyramids**. The primary, ascending sensory pathway carrying spinothalamic fibers will appear throughout these levels as the **medial lemniscus**. Many other more or less distinct fiber tracts course thorough these brain stem levels connecting various parts of the CNS. The nuclei of the cranial nerves are found throughout the pons and medulla, and you can find their approximate location by noting where the nerve exists on the surface. Many other nuclei are scattered up and down the

core of the brain stem from the medulla to the posterior part of the thalamus. These comprise the **reticular formation**, a loosely defined anatomical and functional system.

Your continuing dissection will open the **fourth ventricle** and expose the **cerebellar peduncles** and **transverse fibers of the pons**. Cutting the cerebellum in this way will emphasize its massive fiber connections with the rest of the brain. Continue the dissection noting the fibers of passage and the core nuclei throughout the medulla.

4. Cleaning

Before leaving lab you must properly and carefully store your brain, clean any equipment you used today, and clean your table.

Key Structures:

1. The medial surface of the hypothalamus can be seen on a gross *half-brain cut mid-sagittally*. The following structures will be used to define the limits of the hypothalamus.

anterior commissure
hypothalamic sulcus
lamina terminalis
mammillary bodies
posterior commissure
optic chiasm

2. Locate the hypothalamus on *coronal* and *horizontal* sections of gross brains.

anterior	lateral hypothalamic area
tuberal	medial nuclei (medial preoptic N.,
posterior	dorsomedial N., ventromedial N.,
	paraventricular N.)

3. Locate the following structures on the *ventral surface* of whole brain and *mid-sagittal* cut hemisphere.

amygdala
parahippocampal gyrus
pituitary gland
neurohypophysis
uncus

4. Using *half-brains, frontal and horizontal sections* of a hemisphere identify the following

amygdala (BLA, CeA)
anterior thalamic nucleus
basal nucleus (of Meynert)
body of the fornix
cingulate gyrus
columns of the fornix
entorhinal cortex
fimbria
fusiform gyrus
gyrus rectus
hippocampus (CA3, CA1,
DG)

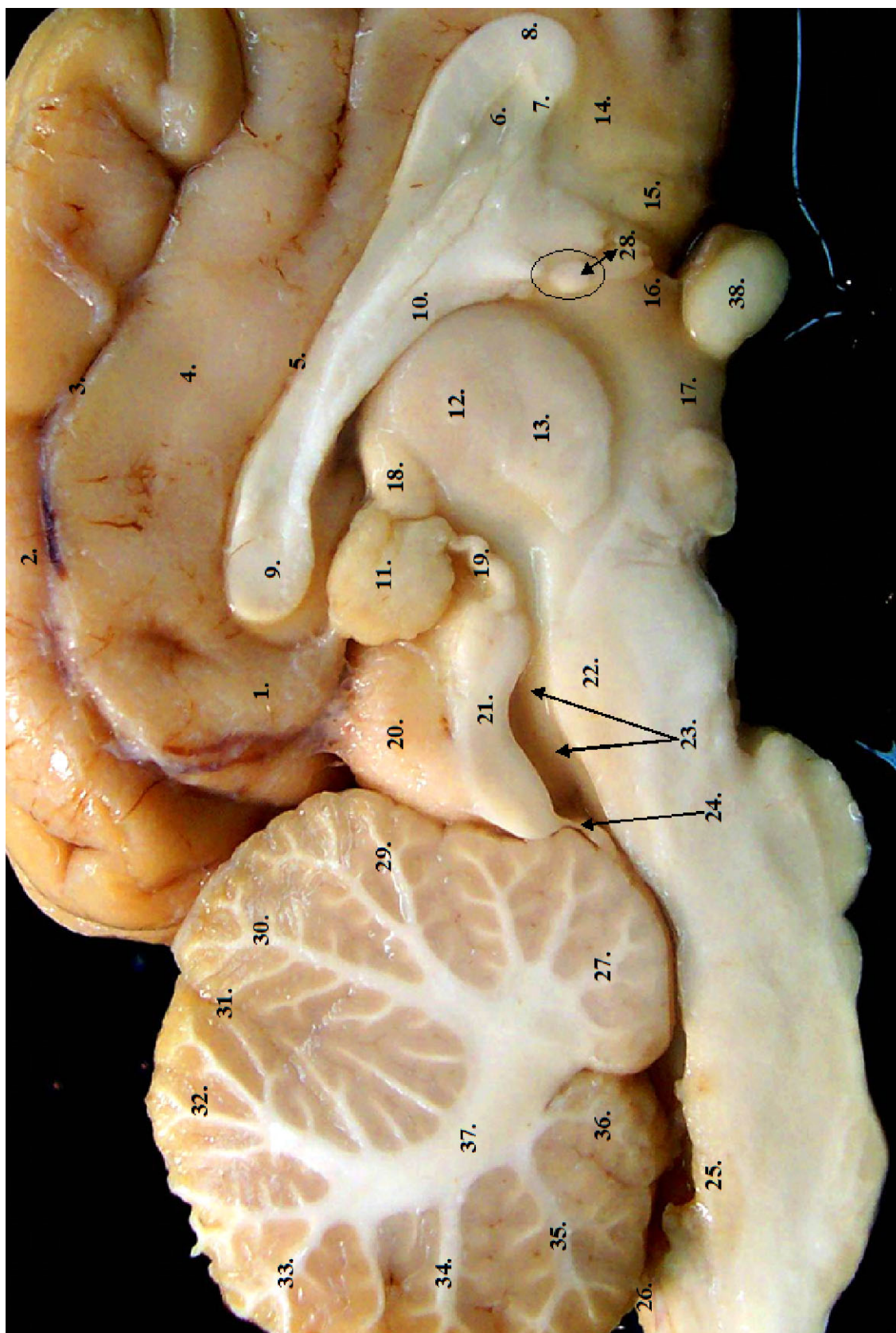
infundibular stalk
mammillary bodies
mammillothalamic tract
nucleus accumbens
optic chiasm
orbital frontal gyrus
parahippocampal gyrus
post-commissural fornix
pre-commissural fornix
septal nuclei
uncus

APPENDIX A: Sheep Brain Photographs

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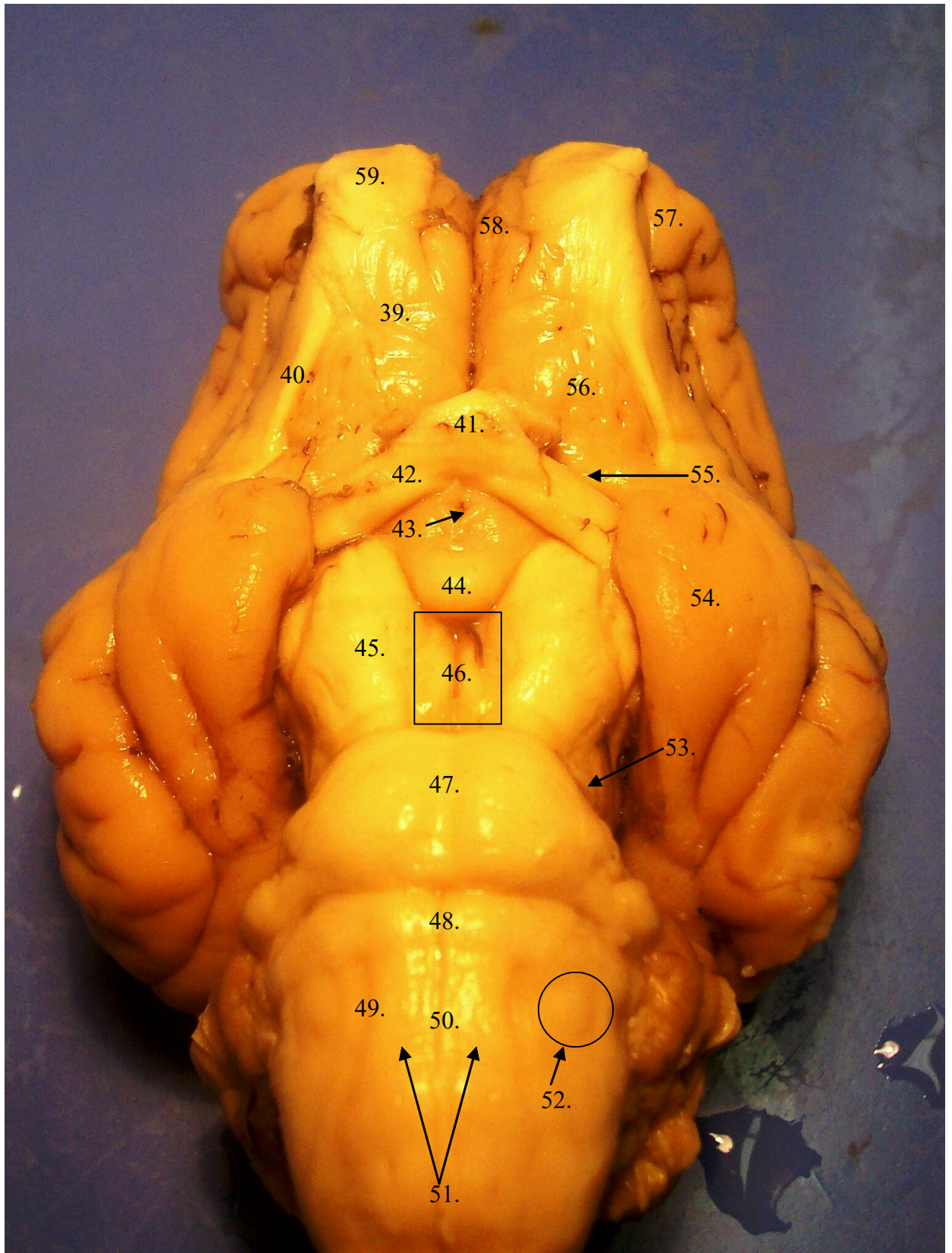
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Mid-sagittal View



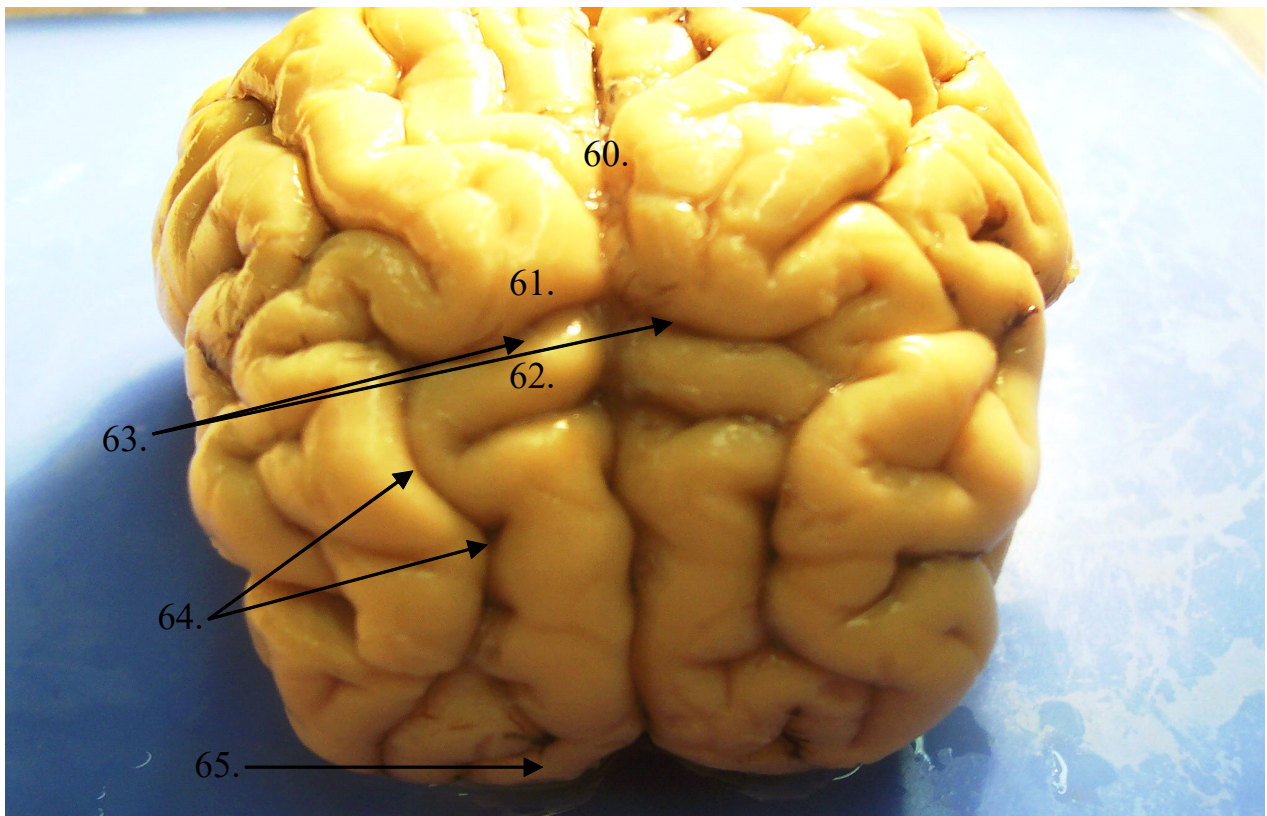
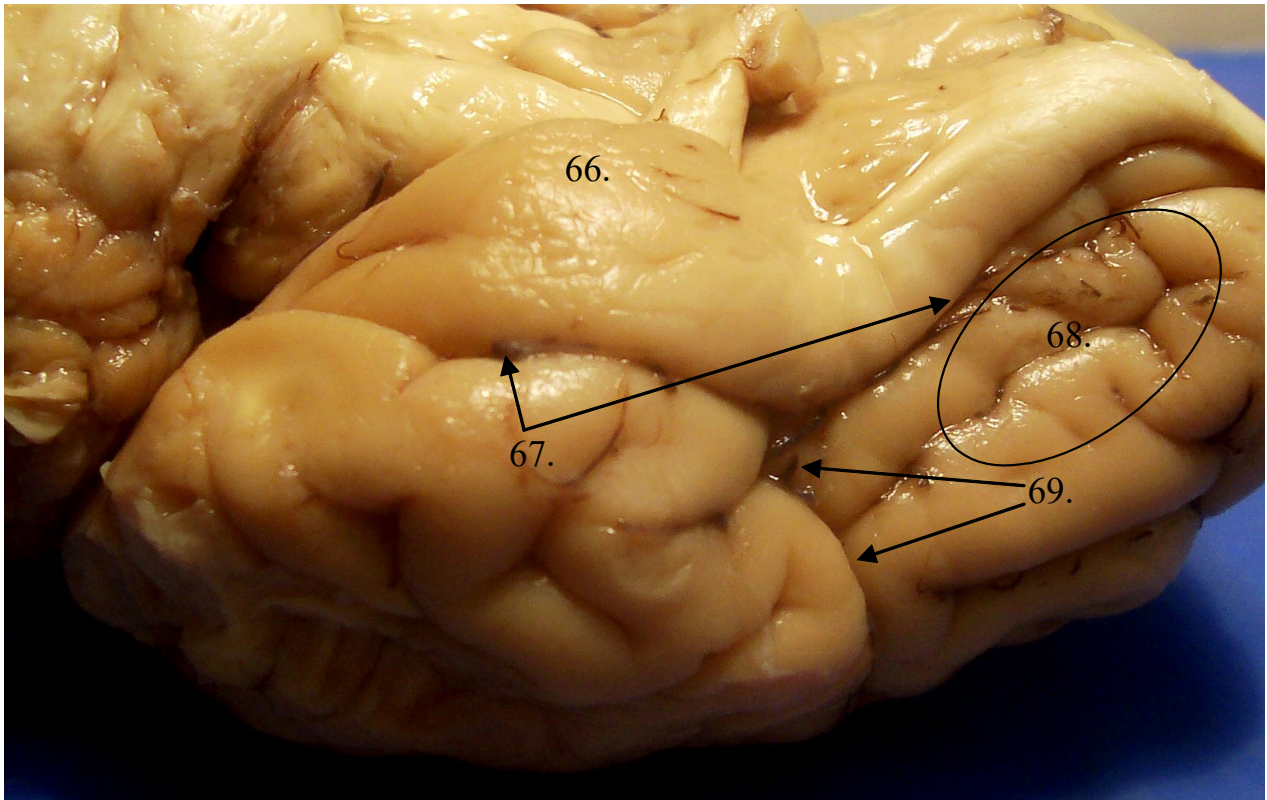
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Ventral View



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Lateral (top) and Anterior (bottom) Views



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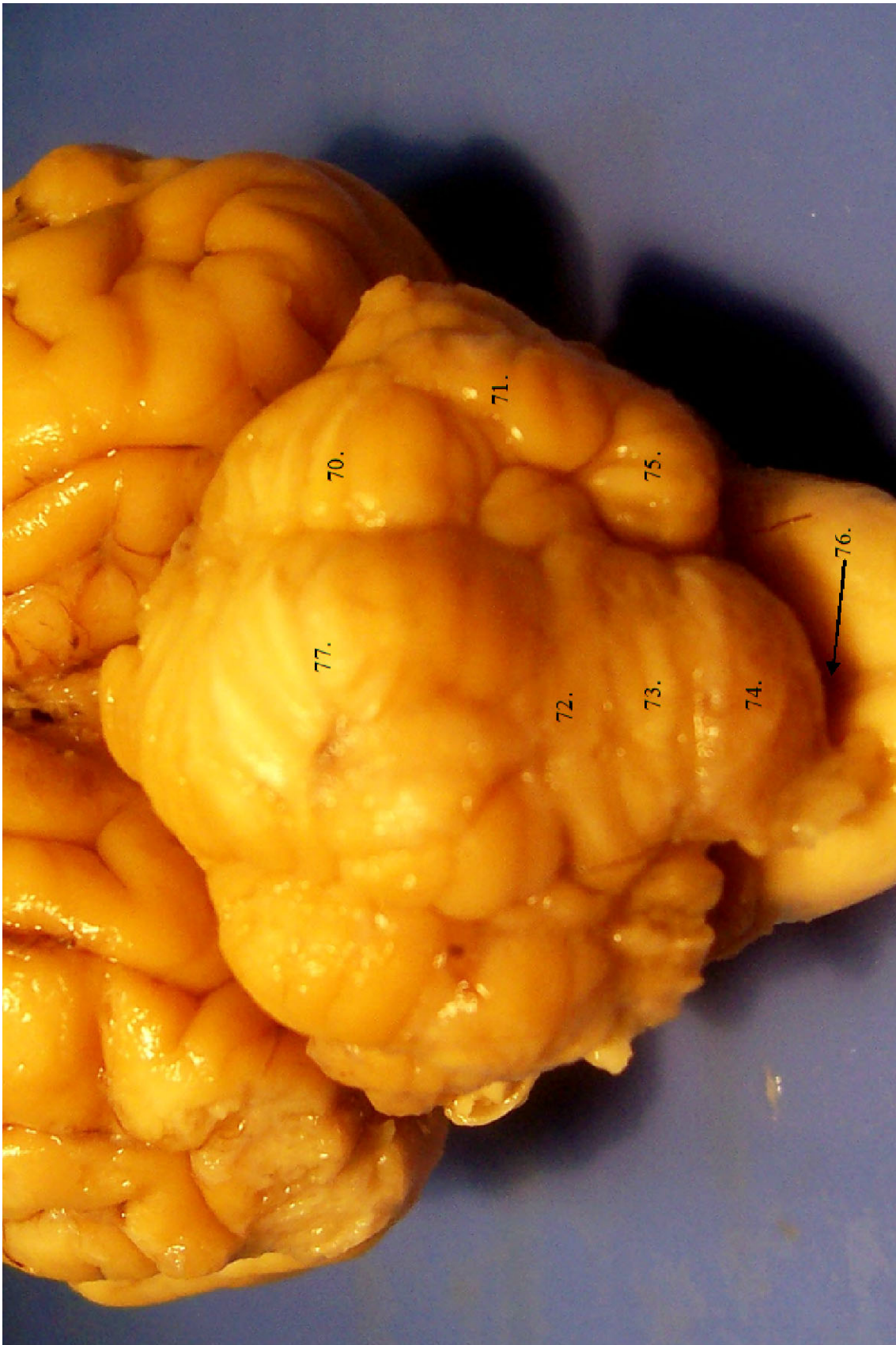
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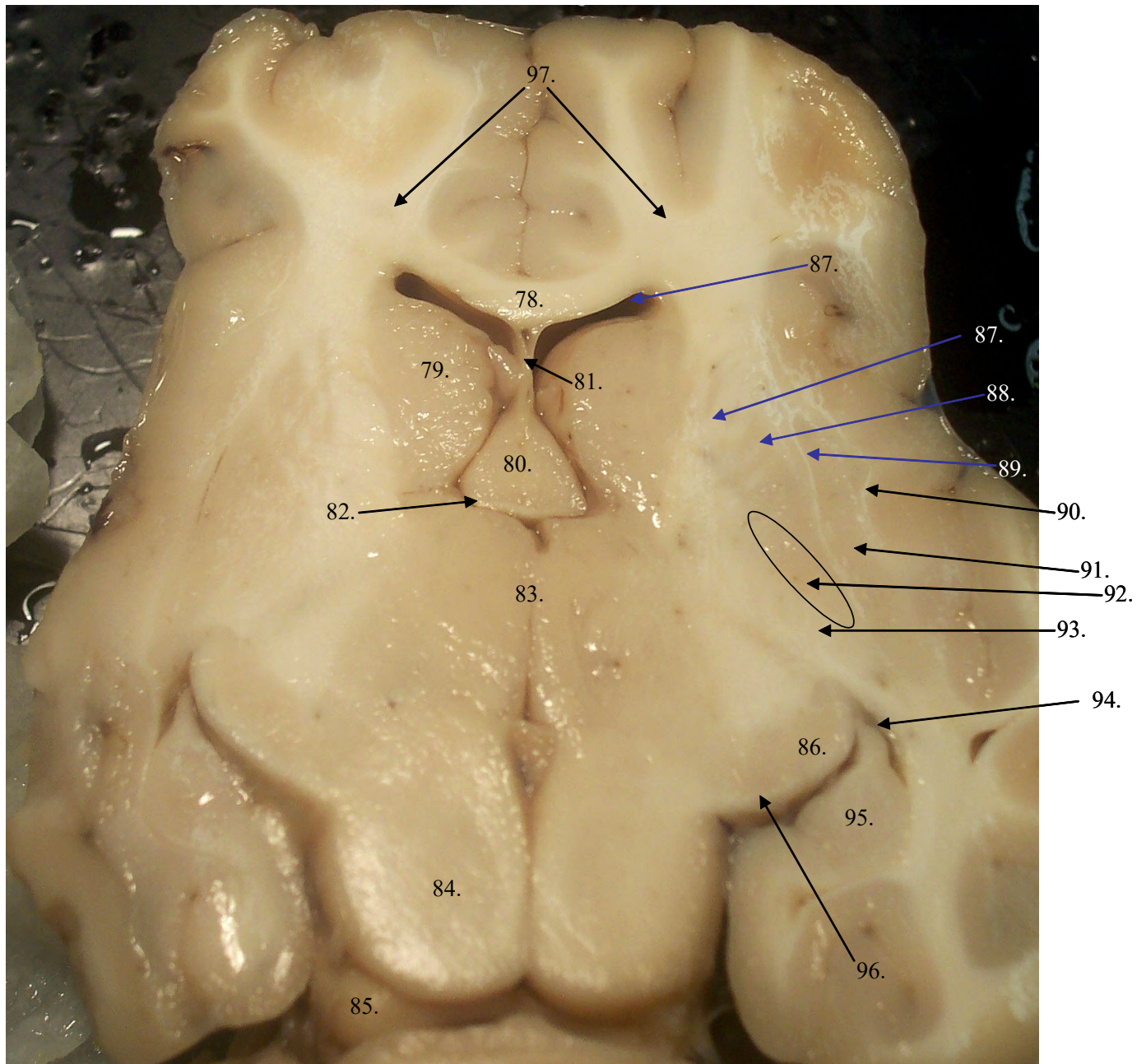
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Cerebellum – Dorsal View



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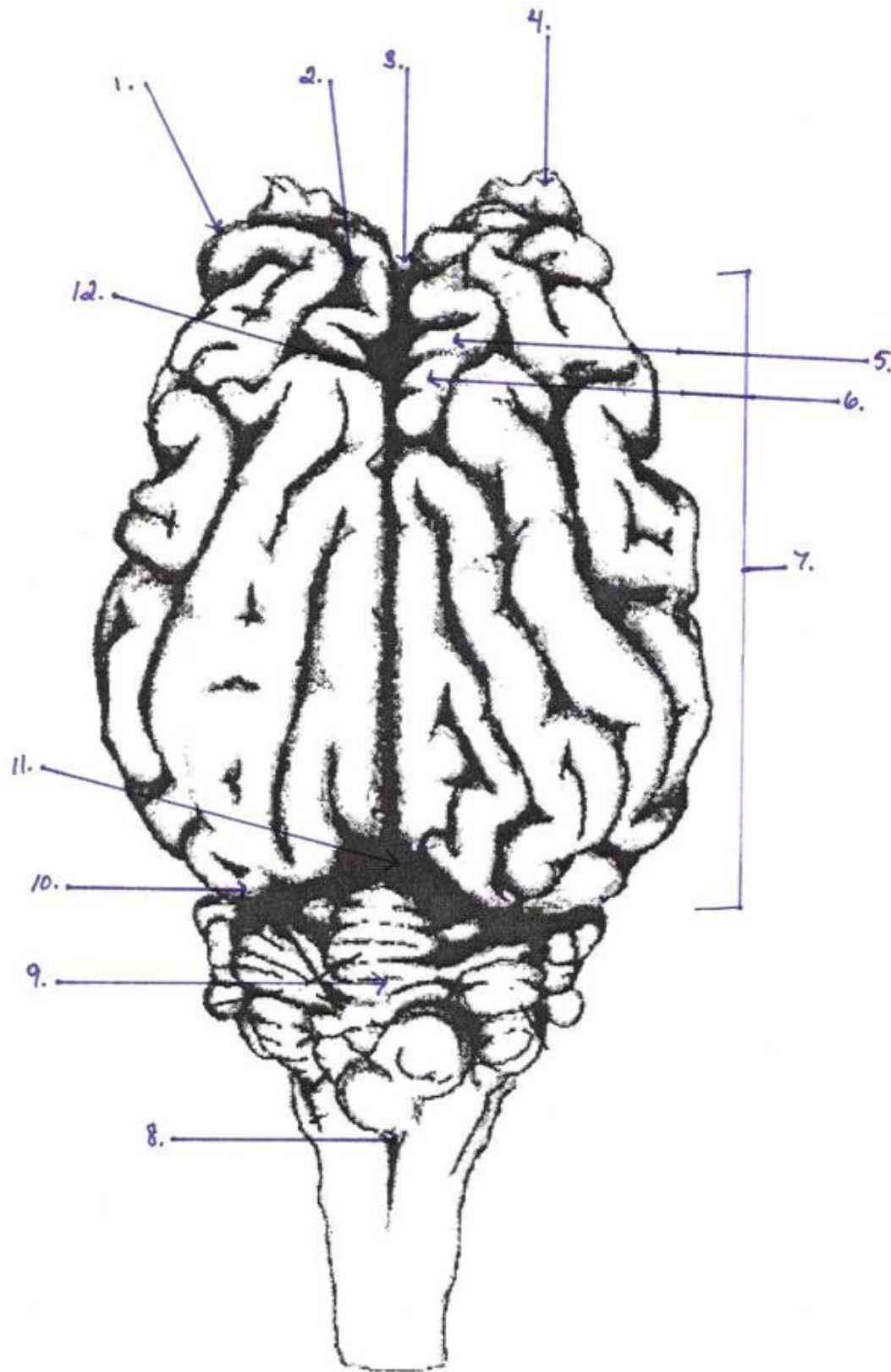
Horizontal Section thru the Dorsal Thalamus



APPENDIX B: Unlabeled Drawings

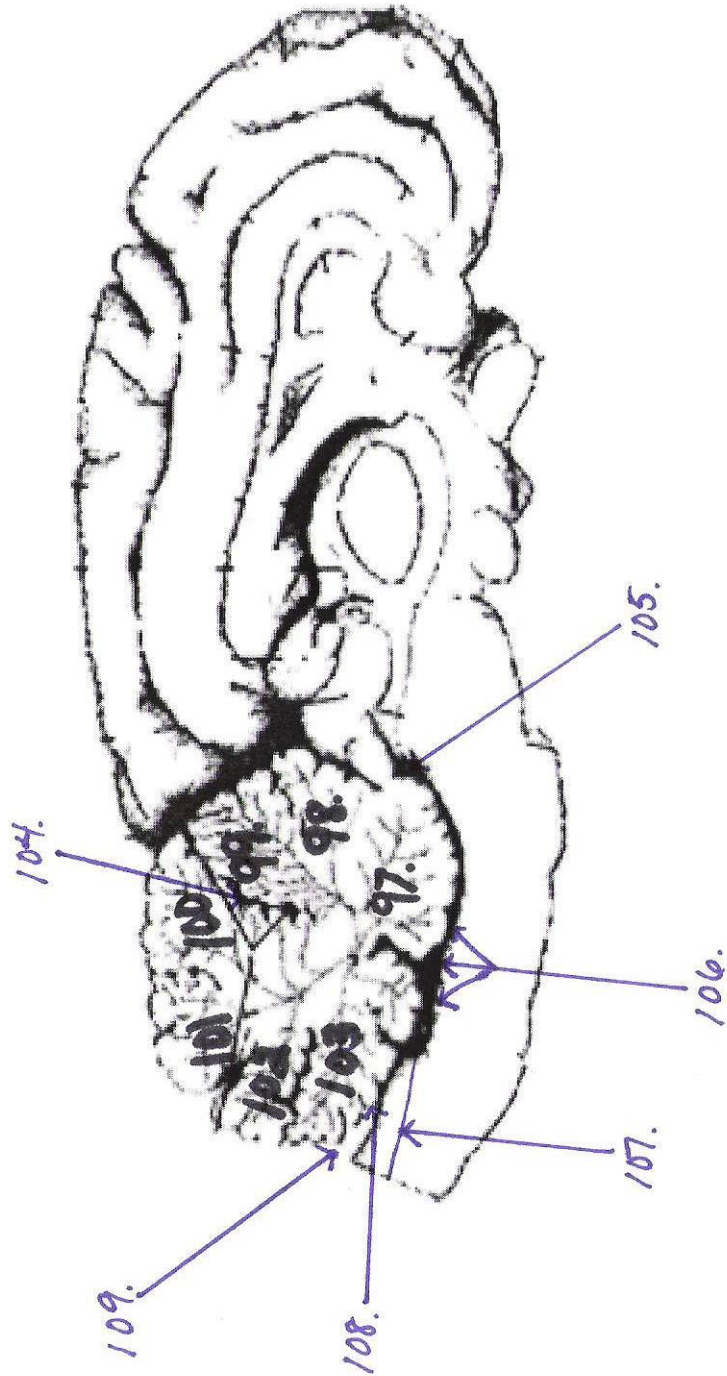
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Dorsal View



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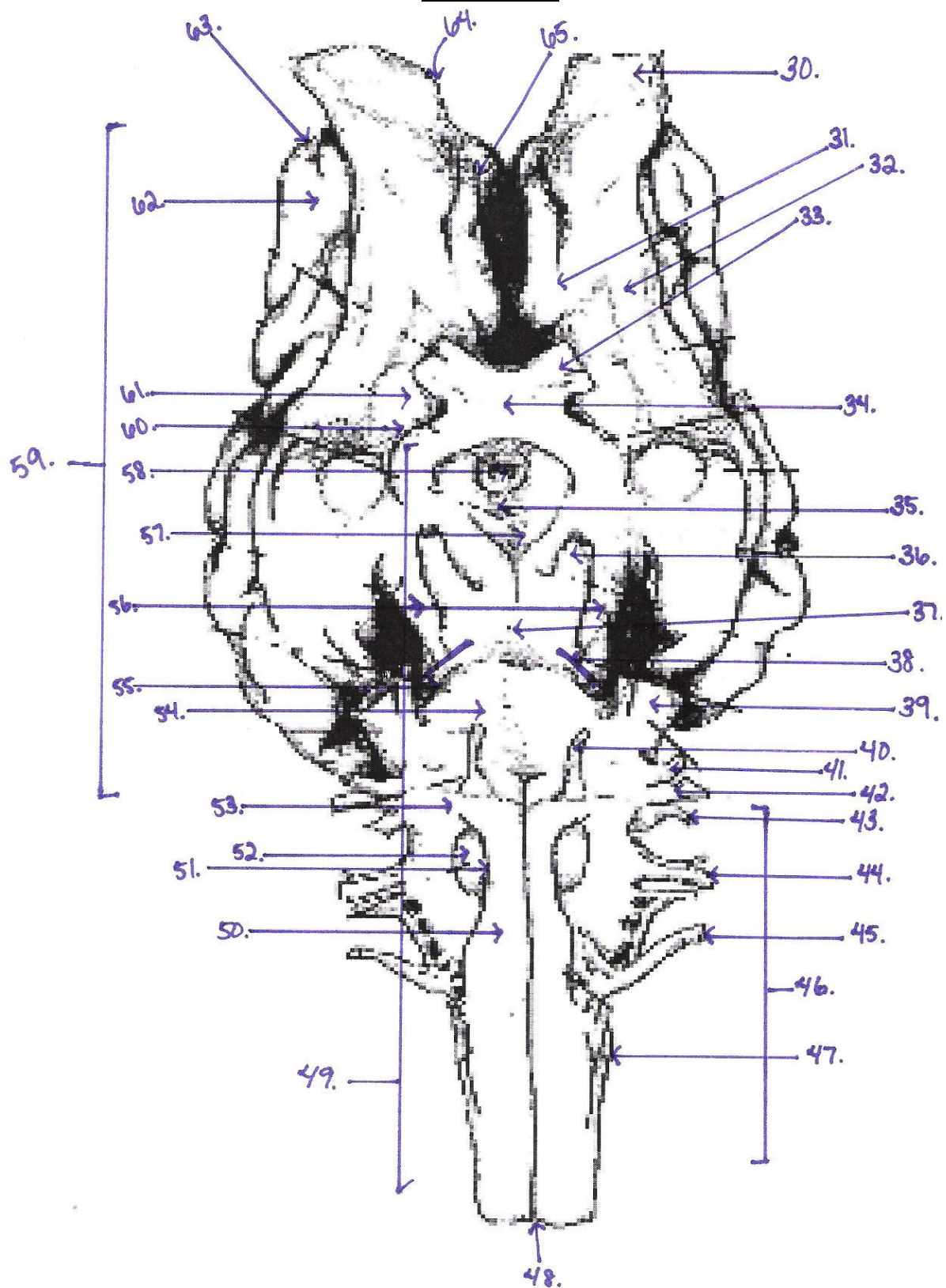
Lateral View



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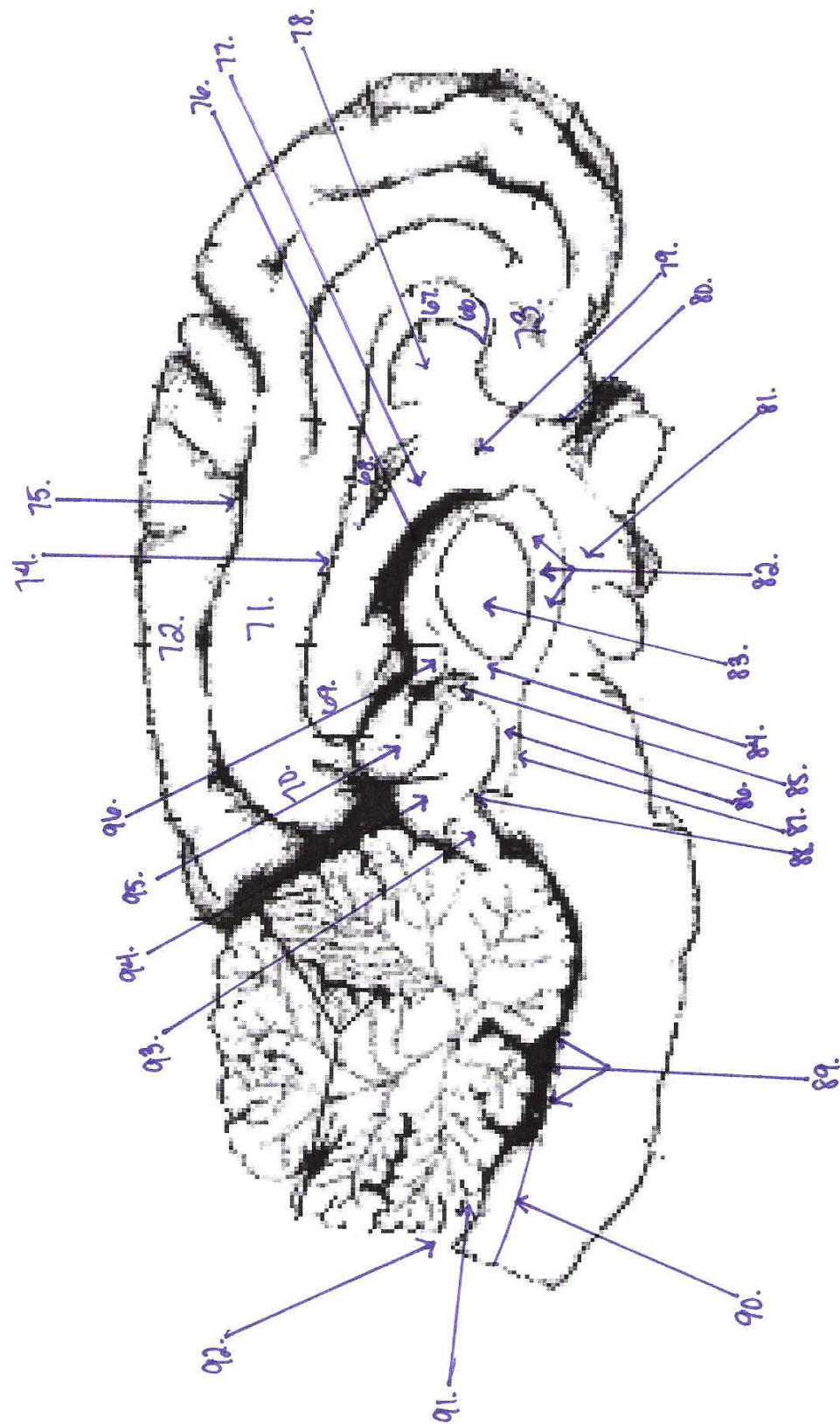
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Ventral View



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Mid-sagittal View



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109. _____

Cerebellum – Mid-sagittal

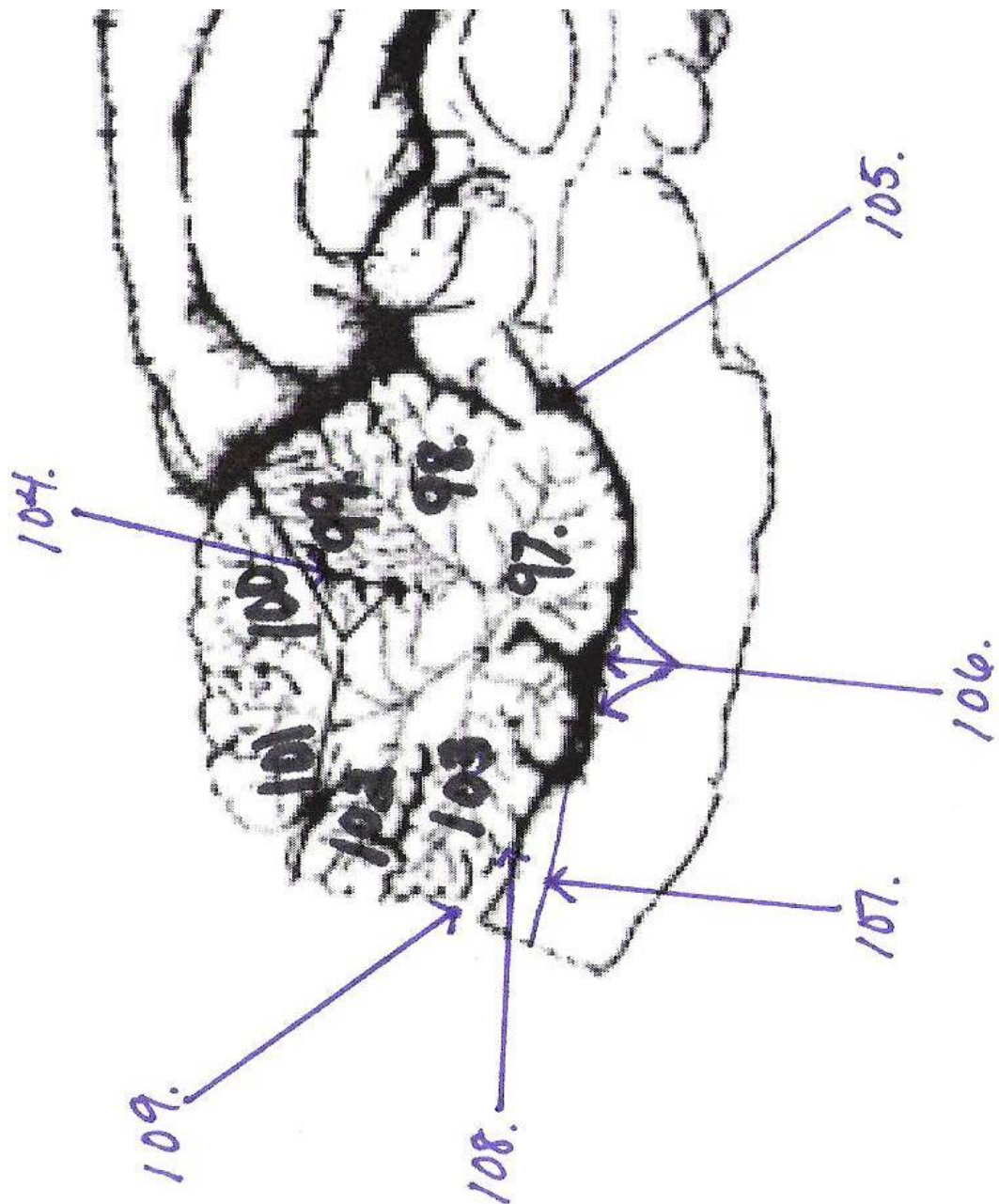


TABLE 1: The Parts of the CNS and Their Associated Cranial Nerves and Cavities

Major Part	Subdivisions	Structures	Cranial Nerves	Cavities
Prosencephalon (forebrain)	Telencephalon	Cerebral cortex Corpora striate Rhinnencephalon	I	Lateral ventricles. Rostral part of third ventricle
	Diencephalon	Thalamus Hypothalamus	II	Major part of third ventricle
Mesencephalon (midbrain)	Mesencephalon	Corpora quadrigemina Cerebral peduncles Tegmentum	III, IV	Cerebral aqueduct
Rhombencephalon (hindbrain)	Metencephalon	Cerebellum Pons	V, VI VII, VIII	Fourth ventricle
	Myelencephalon	Medulla oblongata	IX, X XI, XII	
	Spinal Cord	Spinal cord	Spinal Nerves	Central canal

TABLE 2: The Cranial Nerves

I	Olfactory	Many small fibers arising in olfactory mucous membranes ending in the olfactory bulb	Sensory
II.	Optic	From the retina to the optic chiasm	Sensory
III	Oculomotor	Origin in midbrain; supplies motor control to four muscles of the eye	Motor
IV	Trochlear	Midbrain; motor control to one eye muscle	Motor
V	Trigeminal	Attached to side of pons, very large. Sensory from muscles of mastication, skin of face and scalp, mucous membrane of mouth and nasal cavity, cornea of eye, teeth and dura mater. Motor to muscles of mastication	Mixed
VI	Abducens	Arises from trapezoid body; supplies motor control to one eye muscle	Motor
VII	Facial	Lateral to abducens just behind trigeminal; motor for muscles of expression; sensory from anterior two-thirds of the tongue (taste).	Mixed
VIII	Auditory	Just behind and lateral to facial nerve; serves sense of hearing and equilibrium.	Sensory
IX	Glossopharyngeal	Arises together with vagus nerve behind and ventral to auditory nerve. Sensory from posterior third of tongue (taste), other mouth and throat areas. Motor to salivary gland.	Mixed
X	Vagus	Both sensory and motor for heart, stomach, blood vessels, viscera.	Mixed
XI	Spinal accessory	Runs along lateral surface of medulla and spinal cord receiving fibers along the way. Motor to muscles of neck.	Motor
XII	Hypoglossal	Arises in lower medulla in several more or less distinct roots. Motor to muscles of tongue.	Motor